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Floating-Body Effect Analysis and Multi-Layer Gate Engineering for Vertical-Channel DRAM Access Transistors



2025년 12월

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이 논문을 전자공학석사 학위논문으로 제출함

2026년 01월 09일

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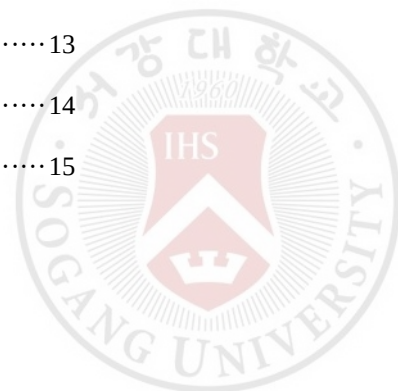
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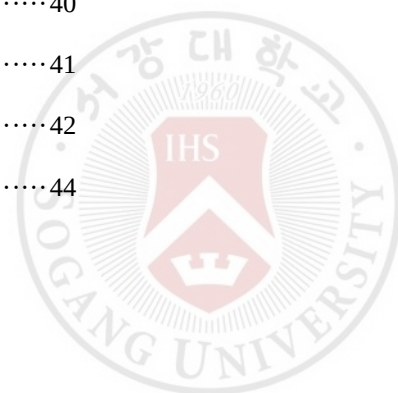


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초록

수직 채널 DRAM 접근 트랜지스터의 부유체 효과 분석 및 다층 게이트 구조 설계

평면형 DRAM의 미세화가 약 $6F^2$ 셀 한계에 근접함에 따라, 산업계는 수직 채널 액세스 트랜지스터(vertical-channel access transistor, VCT)와 개별 저장 커패시터를 결합하여 $4F^2$ 셀을 구현하는 수직 DRAM 구조를 모색하고 있다. 이러한 구조에서 VCT의 실리콘 바디는 주변 절연막에 의해 부분적으로 고립될 수 있으며, 이로 인해 부유체 효과(floating-body effect, FBE)가 발생한다. 게이트 유도 드레인 누설 전류(gate-induced drain leakage, GIDL)와 밴드 간 터널링(band-to-band tunneling, BTBT)에 의해 생성된 정공은 바디에 축적되어 바디 전위를 변화시키고, 결과적으로 데이터 보존 특성을 열화시킨다.

본 연구에서는 보정된 3차원 혼합 모드 TCAD 시뮬레이션을 활용하여 FBE를 정량적으로 분석하고 이를 완화하기 위한 설계 프레임워크를 제시한다. 게이트 길이, 바디 두께, 드레인 측 도핑 농도에 따른 FBE 의존성을 체계적으로 평가한 결과, 짧은 게이트 길이는 드레인 전계 결합을 강화하여 바디 충전을 악화시키는 반면, 얇은 바디는 정전기적 제어를 향상시켜 FBE를 억제하되, 일정 두께 이하에서는 양자 구속 효과(quantum confinement effect, QCE)가 나타나 구동 전류를 제한함을 확인하였다. 데이터 보존 특성과 온 전류 간의 균형을 고려할 때, 약 5 nm의 바디 두께가 최적임을 도출하였다. 또한 드레인 측 도핑을 낮추는 방식은 GIDL을 완화할 수 있으나 액세스 저항을 증가시키므로, 실질적인 FBE 억제를 위해서는 단순한 도핑 감소보다는 전계 공학적 접근이 필요함을 보였다.

이러한 분석을 바탕으로, 채널 양단에는 저 일함수(low-WF) 구간을, 중앙에는 고 일함수(high-WF) 구간을 배치한 3층 일함수(three-layer work-function, 3-layer WF) 게이트 구조를 제안한다. 해당 구조는 채널 방향 전계 분포를 재구성함으로써 두 논리 상태 모두에서 FBE를 동시에 억제하며, 기존 1T-1C DRAM 공정과의 호환성도 유지한다. 본 연구 결과는 맞춤형 게이트 스택 공학을 통해 FBE를 신뢰성 한계 요소가 아닌, 차세대 수직 DRAM 설계를 위한 조절 가능한 설계 파라미터로 전환할 수 있음을 입증한다.

주제어

부유체 효과, 수직 DRAM, 게이트 유도 드레인 누설 전류(GIDL), 양자 구속 효과(QCE), 일함수, 삼중 게이트, TCAD 시뮬레이션, 데이터 보존 특성



Abstract

Floating-Body Effect Analysis and Multi-Layer Gate Engineering for Vertical-Channel DRAM Access Transistors

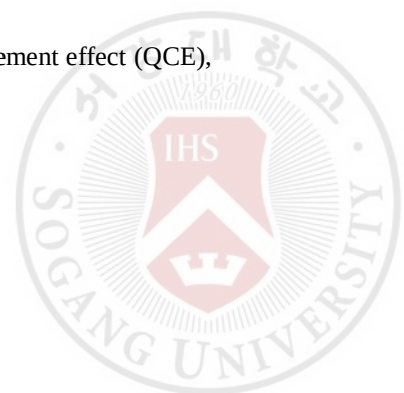
As planar DRAM scaling approached the $\sim 6F^2$ cell limit, the industry has been exploring vertical DRAM architectures that target a $4F^2$ cell by integrating a vertical-channel access transistor (VCT) with a discrete storage capacitor. In this geometry, the silicon body of the VCT can be partially isolated by surrounding dielectrics, giving rise to a floating-body effect (FBE). Holes generated by gate-induced drain leakage (GIDL) and band-to-band tunneling (BTBT) accumulate in the body, shift the body potential, and degrade data retention.

This study establishes a quantitative framework for analyzing and mitigating the FBE using calibrated 3D mixed-mode TCAD. The dependence of FBE on gate length, body thickness, and drain-side doping is systematically assessed: shorter gates strengthen drain-field coupling and exacerbate body charging, whereas thinner bodies improve electrostatic control and suppress FBE until quantum-confinement effects (QCE) emerge and limit drive. Balancing retention and on-current suggests an optimal body thickness of approximately 5 nm. Lowering drain-side doping alleviates GIDL but raises access resistance, indicating that practical suppression requires electric-field engineering rather than simple doping reduction.

Building on these insights, a three-layer work-function (3-layer WF) gate—low-WF segments at both channel ends and a high-WF center—is proposed to redistribute the longitudinal field and concurrently mitigate FBE in both logic states, while remaining compatible with standard 1T–1C DRAM processing. The results demonstrate that tailored gate-stack engineering can transform FBE from a reliability constraint into a tunable design parameter for next-generation vertical DRAM.

KEYWORDS

Floating-body effect, Vertical DRAM, Gate-induced drain leakage (GIDL), Quantum confinement effect (QCE), Work-function, Three-layer gate, TCAD simulation, Retention characteristics



Chapter 1. Introduction

1.1 Scaling Limitations and Emergence of Floating-Body Effects in Vertical DRAM

Continuous scaling of one-transistor–one-capacitor (1T1C) dynamic random-access memory (DRAM) cells leads to severe degradation in retention time and parasitic coupling [1.1]–[1.5]. Vertical or trench-type capacitors have been introduced to maintain sufficient storage capacitance, but these structures still rely on planar-channel access transistors and cannot reduce the footprint below $6F^2$, where F denotes the minimum lithographic feature size [6]–[9]. To overcome this limitation, vertical integration concepts such as vertical-channel transistor (VCT) DRAM and vertical-stacked (VS) DRAM have been proposed, where the cell components are arranged along the vertical axis to further reduce the cell area beyond the planar $6F^2$ limit. Figure 1.1 summarizes this technological evolution, highlighting the transition from conventional 2D cell layouts toward 3D vertical-channel configurations enabled by CMOS wafer bonding and stacked array wafers [1.10]. Prior to the advent of 3D-stackable DRAM, the $4F^2$ vertical-channel DRAM was proposed as an interim architecture for continued cell scaling within conventional DRAM process limits, since true 3D stacking requires complex processes (e.g., wafer bonding, through-silicon-via (TSV) formation, precise layer alignment) that are difficult to integrate into standard DRAM fabrication flows [1.11]–[1.13].

Although the vertical architecture improves scalability, it simultaneously leaves the silicon body partially isolated by surrounding dielectrics, effectively forming a floating body [1.9], [1.14]. Under high drain-field conditions, holes generated near the drain by gate-induced drain leakage (GIDL) and band-to-band tunneling (BTBT) are stored in the isolated body, shifting the body potential [1.15]–[1.17]. Notably, even in fully depleted bodies, a floating-body effect can arise under non-equilibrium operation when excess holes generated by BTBT or GIDL cannot be efficiently discharged, leading to transient body-potential modulation. Figure 1.2 schematically illustrates this hole-accumulation-induced body charging mechanism: holes produced at the drain-side junction are trapped in the body region beneath the gate and progressively charge the body. The floating-body effect (FBE) increases the body potential (manifested as downward band bending) and weakens electrostatic depletion, which shifts the effective threshold voltage. As a result, the device suffers from threshold-voltage instability, read-disturb, and retention degradation [1.18].



DRAM TECHNOLOGY EVOLUTION FROM 2D SCALING TO 3D ARCHITECTURES

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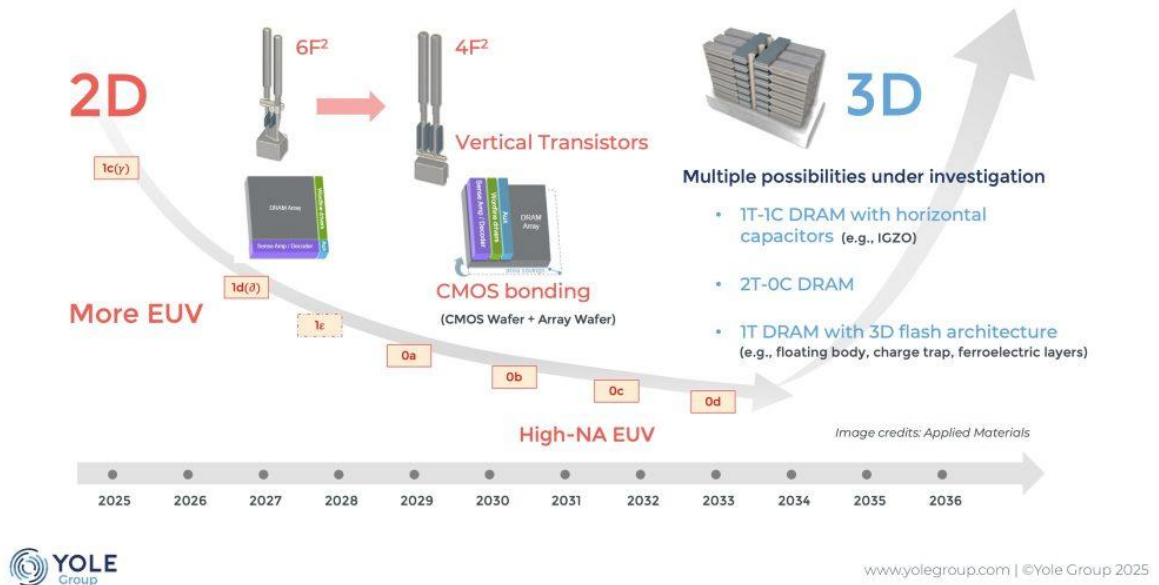


Figure 1.1. Evolution of DRAM technology from 2D planar scaling toward 3D vertical architectures [1.11].

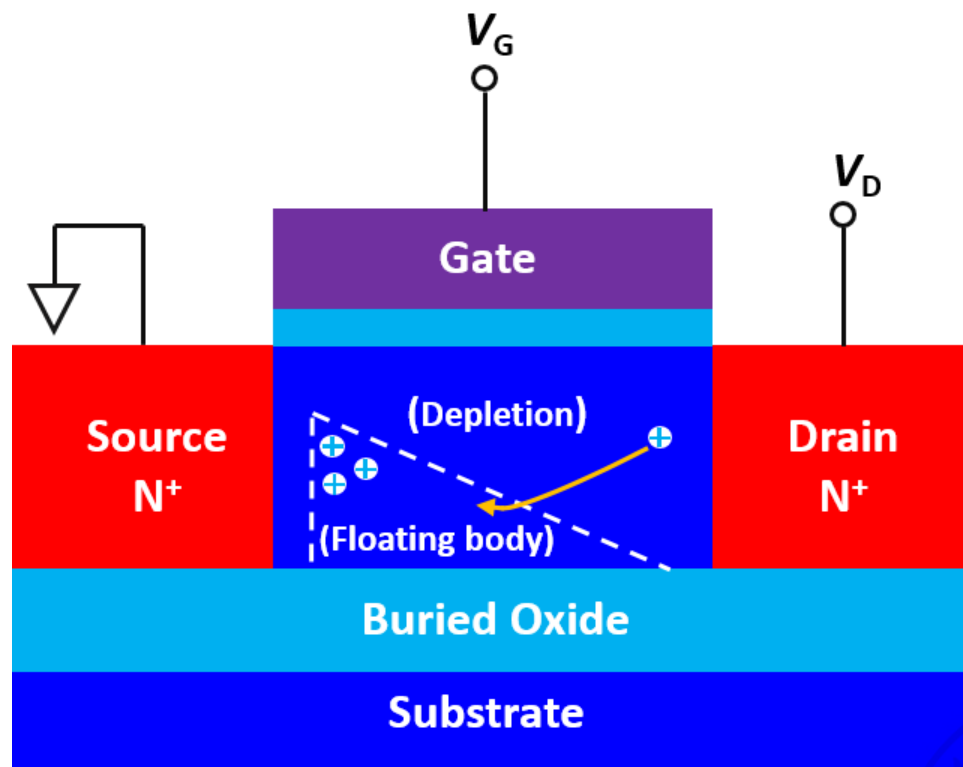


Figure 1.2. Schematic illustration of the floating-body effect (FBE).

1.2 Research Motivation and Scope

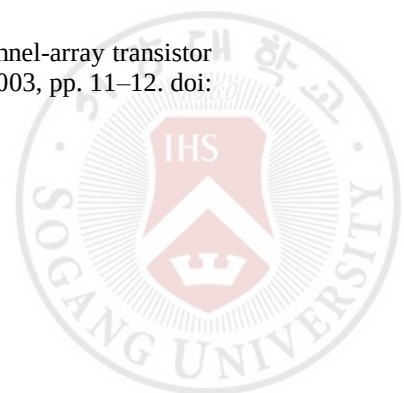
In this study, we focus on the nanowire-type vertical-channel transistor (VCT) DRAM architecture, where strong geometrical confinement and an electrically floating body make the device particularly susceptible to floating-body effects under non-equilibrium operation.

The aim of this work is to analyze and quantify the impact of the FBE on data retention in vertical DRAM cells. Through detailed TCAD simulations, the dependence of FBE on key design parameters—such as gate length, body thickness, and drain doping concentration—is systematically investigated. Special attention is paid to the variations in body potential and retention characteristics between the ‘0’ and ‘1’ states under both static and dynamic conditions.

The insights obtained from this analysis form the foundation for developing a structural approach to mitigate FBE through drain-field modulation and gate stack optimization, which will be discussed in the following chapters. By clarifying the physical mechanisms and scaling trends of the FBE, this work aims to provide design guidelines for reliable and scalable vertical DRAM technologies.

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Chapter 2. Impact of Floating-Body Effect on DRAM Operation

2.1 Device Structure and Simulation Setup

2.1.1 Model Calibration Using Reference VCT Device

To quantitatively investigate the influence of the FBE on data retention, a vertical-channel transistor (VCT) representative of a vertical DRAM cell was implemented in Synopsys Sentaurus TCAD simulation [2.1]. The physical parameters were first calibrated to experimental data from a previously reported pillar-type VCT [2.2], as verified by the I_D - V_G and I_D - V_D fitting shown in Figure 2.1. Because the reference device has relatively large dimensions, it primarily serves as a calibration benchmark. The geometry was subsequently rescaled to DRAM-relevant dimensions, where gate-induced drain leakage (GIDL) and band-to-band tunneling (BTBT) become dominant contributors to the FBE.

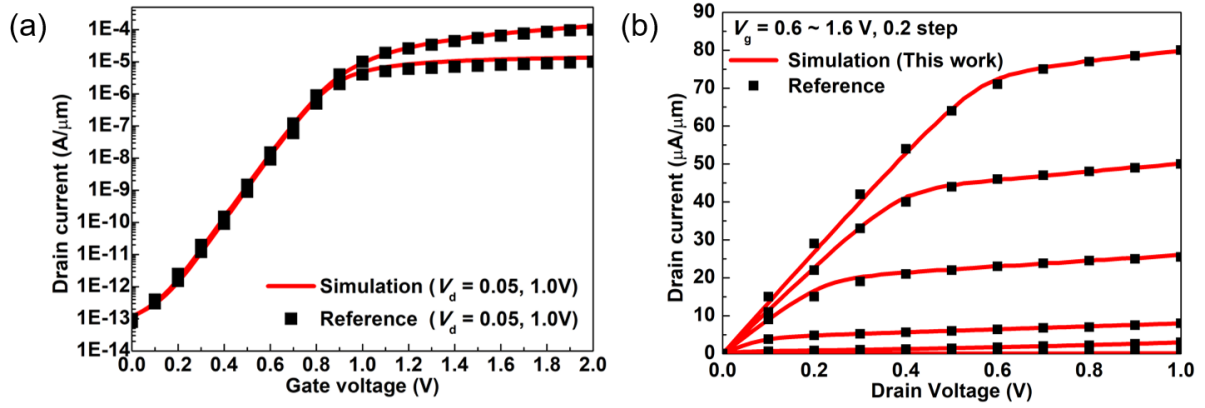
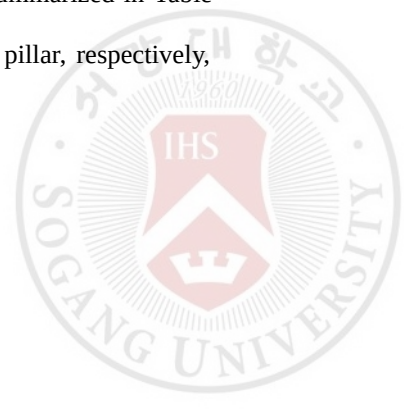


Figure 2.1. Comparison of simulated and measured characteristics for the reference pillar-type vertical-channel transistor (VCT) used for TCAD calibration. (a) Transfer curve (I_D - V_G) and (b) output curve (I_D - V_D) [2.2].

2.1.2 Geometry and Key Design Parameters

The rescaled VCT structure used in this study is schematically illustrated in Figure 2.2. The device consists of a silicon pillar with a gate-all-around (GAA) stack, heavily doped source/drain regions, and a partially isolated body. Holes generated by GIDL can accumulate in this floating body, increasing body potential and thereby modulating the threshold voltage [2.3]-[2.5]. The structural and electrical parameters are summarized in Table 2.1. The bit line (BL) and storage node (SN) are connected to the top and bottom of the pillar, respectively, forming the access transistor of a vertical DRAM cell.



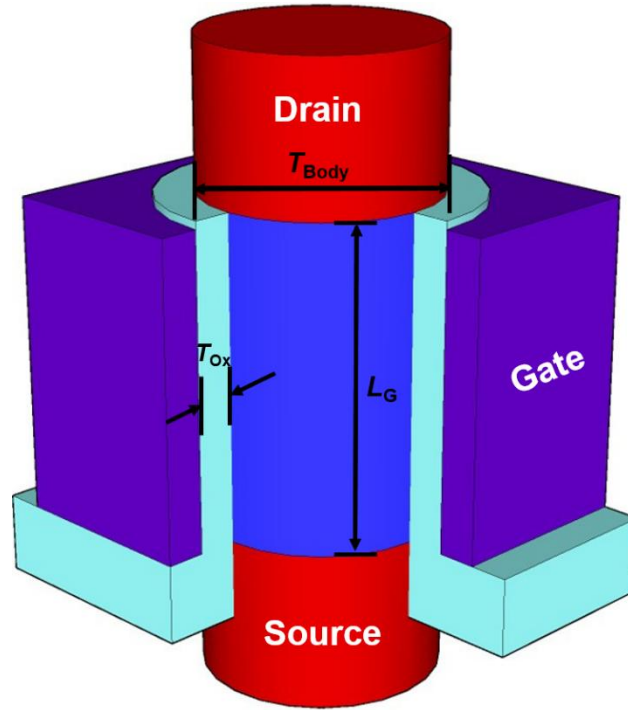


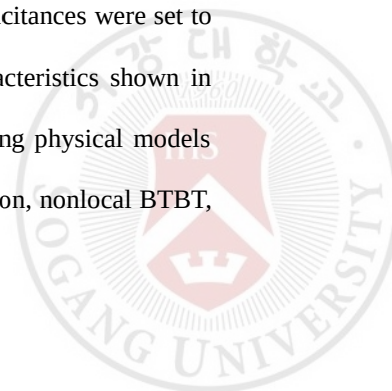
Figure 2.2. 3D schematic of the simulated vertical-channel DRAM cell used for FBE analysis.

Symbol	Parameter	Values
L_G	Gate length	60 nm
T_{Ox}	Spacer thickness	1 nm
T_{Body}	Channel width	12 nm
N_{SD}	Source/Drain doping concentration	$1 \times 10^{20} \text{ cm}^{-3}$
N_{Ch}	Channel doping concentration	$1 \times 10^{17} \text{ cm}^{-3}$

Table 2.1. Key design parameters used in the Sentaurus TCAD simulation of the vertical-channel transistor (VCT) for FBE analysis.

2.1.3 Simulation Framework and Physical Models

A 3D mixed-mode TCAD simulation framework was employed to capture the interaction between the transistor and peripheral nodes during DRAM operation. The storage-node and bit-line capacitances were set to 10 fF and 100 fF, respectively, to emulate array conditions. The simulated transfer characteristics shown in Figure 2.3 confirm proper electrostatic control and nominal device operation. The following physical models were enabled: phonon-limited mobility (PhuMob), Shockley–Read–Hall (SRH) recombination, nonlocal BTBT,



and high-field velocity saturation. Time-domain TCAD simulations were conducted under bias conditions corresponding to dynamic and static retention modes, enabling a quantitative analysis of body-potential evolution and data retention for both ‘1’ and ‘0’ states.

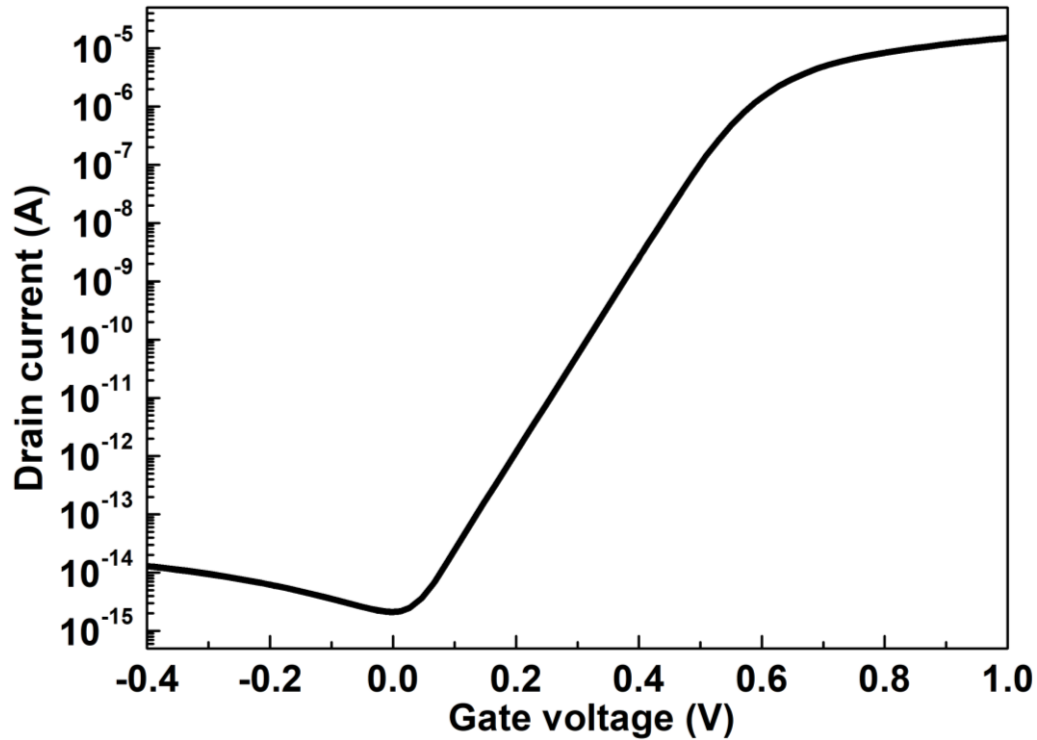
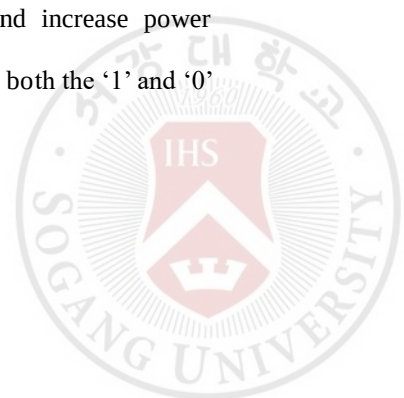


Figure 2.3. Simulated transfer characteristics (I_D - V_G) of the vertical-channel DRAM access transistor at $V_D=1.0V$

2.2 Impact of Floating-Body Effect on Dynamic Retention

2.2.1 Introduction

The FBE is a critical reliability concern for vertical-channel transistors because the partially isolated body can store excess holes generated during DRAM operation. Dynamic retention refers to the data retention behavior when the WL is turned off while the BL is toggling. As the BL voltage (V_{BL}) toggles, holes are generated and stored in the floating body, while the storage capacitor loses charge and the SN voltage (V_{SN}) shifts. Among various failure modes, the dynamic retention characteristic is most directly influenced by FBE: repeated refresh/access operations perturb the body potential and can accelerate charge loss and increase power consumption. To clarify these mechanisms, the dynamic retention behavior was analyzed for both the ‘1’ and ‘0’ storage states.



2.2.2 '1' State Operation

Figure 2.4 illustrates the case where the SN holds a '1' state. Figure 2.4(a) shows the transient bias waveform; Figures 2.4(b) and 2.4(c) depict the corresponding band diagrams during the BL-high and BL-low intervals, respectively.

When $BL = 1$ V, dual-sided GIDL occurs at both the SN and BL junctions, injecting holes into the floating body. The accumulated holes increase body potential as shown in Figure 2.4(b). When the BL voltage returns to 0 V, these holes discharge predominantly toward the BL side, causing the body potential to decrease accordingly, as shown in Figure 2.4(c). The resulting change in body potential alters the channel barrier, driving electrons from the BL toward the SN and causing a V_{SN} reduction. This periodic accumulation/discharge of holes gradually degrades the stored charge, thereby limiting the dynamic retention time in the '1' state.

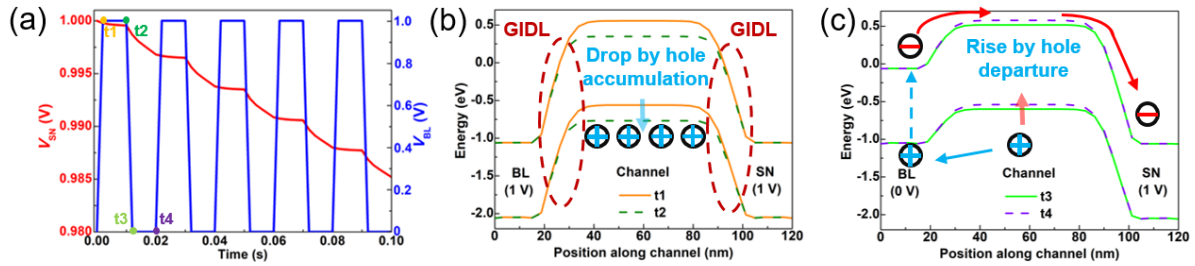


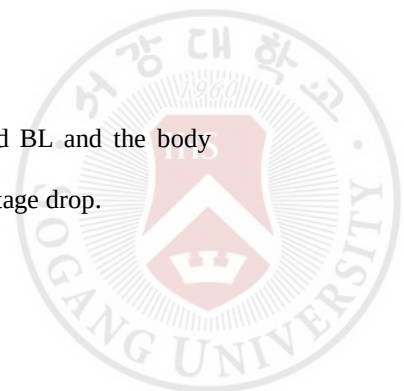
Figure 2.4. Dynamic retention behavior of the vertical-channel DRAM cell when the storage node (SN) stores a '1' state. (a) Simulated timing diagram of bit-line (BL) and SN voltages used for transient analysis. (b) Energy-band configuration under BL-high bias. (c) Energy-band configuration under BL-low bias.

2.2.3 '0' State Operation

Figure 2.5 summarizes the dynamic retention when the SN initially stores a '0' state. The response is not a simple inversion of the '1' case because the balance between FBE and GIDL generation differs.

Under $BL = 1$ V, GIDL is pronounced on the BL side and injects holes into the body. Although the accumulated charge is smaller than in the '1' state (one dominant junction), the body exhibits body potential increasing, which lowers the channel barrier near the SN/body junction and raises the SN voltage during the BL-high period, as shown in Figure 2.5(b).

When BL returns to 0 V as shown in Figure 2.5(c), the stored holes discharge toward BL and the body potential recovers; electrons near the BL side flow toward the SN, producing a small SN-voltage drop.



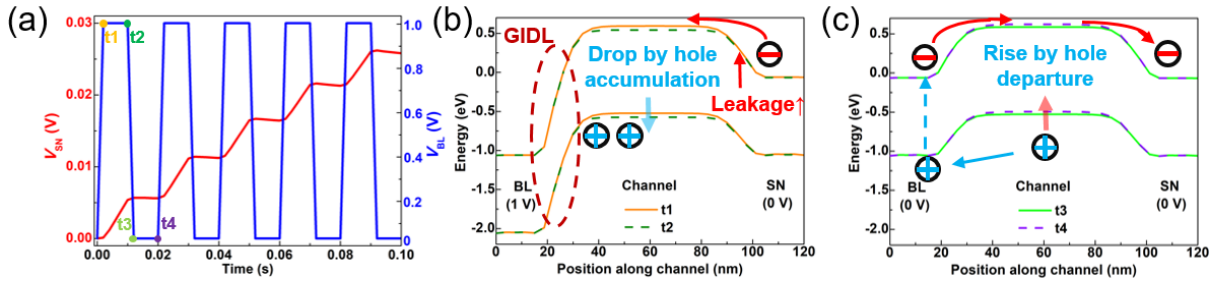


Figure 2.5. Dynamic retention behavior of the vertical-channel DRAM cell when the storage node (SN) stores a ‘0’ state. (a) Simulated BL and SN voltage waveforms for transient operation. (b) Energy-band configuration during BL-high bias. (c) Energy-band configuration during BL-low.

2.3 Impact of Floating-Body Effect on Static Retention

2.3.1 Introduction

Static retention refers to the data retention behavior when the WL is turned off and the BL is precharged to $V_{DD}/2$ (0.5 V). In addition to dynamic operation, static retention is another critical factor determining the reliability of DRAM cells. While dynamic retention reflects charge loss during periodic access, static retention represents the cell’s ability to preserve data with the word line (WL) off. In practical DRAM circuits, this corresponds to the condition where BL is precharged to $V_{DD}/2$ and the cell must retain the stored charge in the SN for a given hold period. Even under such static bias, the FBE can significantly influence long-term stability by modifying the leakage paths through the body region.

2.3.2 ‘1’ State Operation

Figure 2.6 shows the static retention characteristics when the SN stores a ‘1’ state. In this simulation, the WL was off, the SN was initialized to 1 V, and the BL was held at 0.5 V to emulate the half-precharge condition. As shown in Figure 2.6(a), V_{SN} gradually decreases due to the GIDL arising at the SN junction. The energy-band diagram in Figure 2.6(b) indicates that dual-sided GIDL at both junctions injects holes into the body, causing the body potential to increase. The resulting self-limiting band bending suppresses further GIDL on both sides, thereby improving static retention for the ‘1’ state.



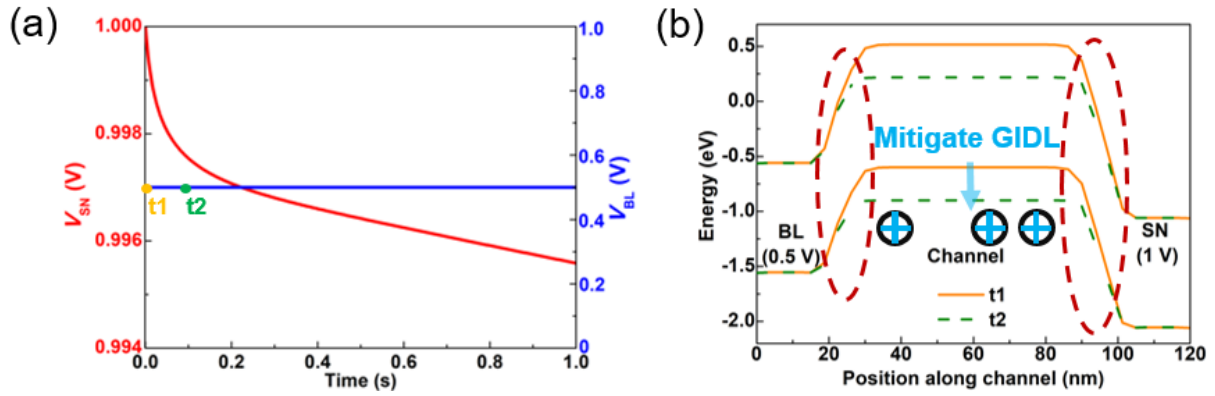


Figure 2.6. Static retention characteristics of the vertical-channel DRAM cell when the storage node (SN) stores a ‘1’ state. (a) Simulated SN voltage variation over time under the half-precharge condition ($V_{BL}=0.5$ V, $V_{SN}=1.0$ V). (b) Corresponding energy-band diagram showing that dual-side gate-induced drain leakage (GIDL) generates holes in the floating body.

2.3.3 ‘0’ State Operation

Figure 2.7 presents the static retention behavior of the vertical-channel DRAM cell under the ‘0’ state. As shown in Figure 2.7(a), V_{SN} increases more noticeably than the voltage drop observed in the ‘1’ state. This behavior arises because, while the ‘1’ state is mainly affected by a relatively small electron-injection leakage from the BL side ($V_{BL} = 0.5$ V), the intrinsic SN-to-body leakage in the ‘0’ state is larger, leading to a greater net disturbance during retention.

The band diagrams in Figure 2.7(b) provide further insight. Since the BL-side GIDL is very weak at $V_{BL} = 0.5$ V, little hole generation occurs, and the body potential remains nearly unchanged. With minimal body modulation, the barrier near the SN junction is almost constant; however, the intrinsic SN leakage still causes a gradual rise of V_{SN} , producing the larger excursion seen in Figure 2.7(a). Thus, despite the weak GIDL, the ‘0’ state does not benefit from improved retention stability, and its disturbance is dominated by SN-side leakage rather than FBE-driven body-potential fluctuations.

However, if BL-side GIDL becomes sufficiently strong—such as under higher junction fields or scaled geometries—the generated holes can raise the body potential appreciably, lowering the SN-to-body barrier. In this case, the increased hole accumulation would amplify the SN-side leakage, causing V_{SN} to rise even faster. This indicates that a stronger FBE can, in principle, worsen the ‘0’-state retention by indirectly enhancing the SN leakage through body-potential modulation.

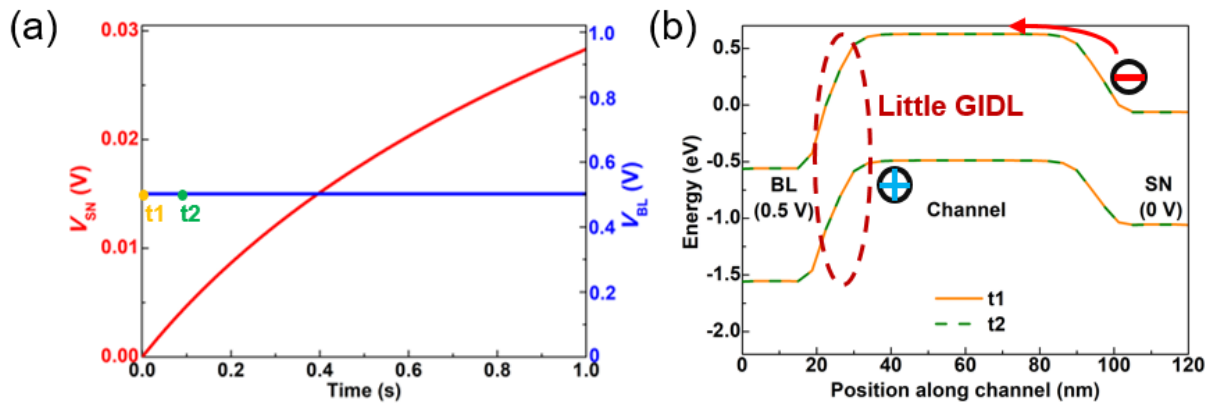


Figure 2.7. Static retention characteristics of the vertical-channel DRAM cell when the storage node (SN) stores a ‘0’ state. (a) Simulated SN voltage evolution under the same half-precharge condition ($V_{BL} = 0.5V$). (b) Energy-band configuration illustrating that, although BL-side GIDL is mitigated, the increased body potential reduces the barrier near the SN junction and increases SN leakage.

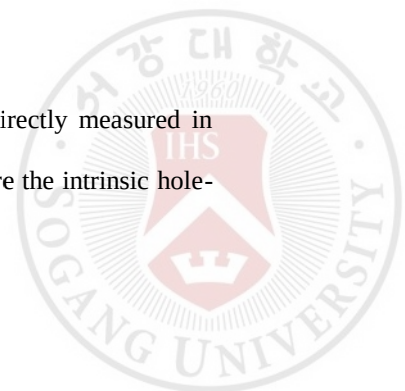
2.4 Evaluation of Pure Floating-Body Effect

2.4.1 Introduction

In both dynamic and static retention operations, the variation of V_{SN} arises from a mixture of floating body-induced body-potential modulation and junction-field-driven leakage currents at the BL and SN junctions. Because these two mechanisms occur simultaneously and influence the V_{SN} on similar time scales, the V_{SN} waveform alone cannot reveal how strongly the floating body itself is being charged. A direct quantification of the intrinsic FBE therefore, requires a metric that separates body-potential modulation from conventional leakage processes.

To address this issue, the body-potential variation (BPV) during the BL-high phase ($V_{BL} = 1$ V) is used as a leakage-independent indicator of FBE strength. BPV represents the rise in body potential that results solely from hole accumulation inside the partially depleted body. As illustrated in Figure 2.8, the difference between the t_1 and t_2 energy-band profiles directly corresponds to the degree of body charging, making BPV the most physical and intuitive measure of FBE strength. In the ‘1’ state, GIDL occurs at both the BL and SN junctions, generating holes from both sides and resulting in a larger BPV. In contrast, the ‘0’ state produces holes only at the BL-side junction, so the overall hole accumulation is smaller and the BPV remains limited.

Since the floating body has no accessible electrical terminal, its potential cannot be directly measured in fabricated devices. As a result, BPV can only be evaluated through TCAD simulation, where the intrinsic hole-



accumulation process can be isolated from parasitic leakage and external measurement constraints. This approach provides a clear and device-physics-driven metric for assessing the pure FBE without contamination from junction leakage.

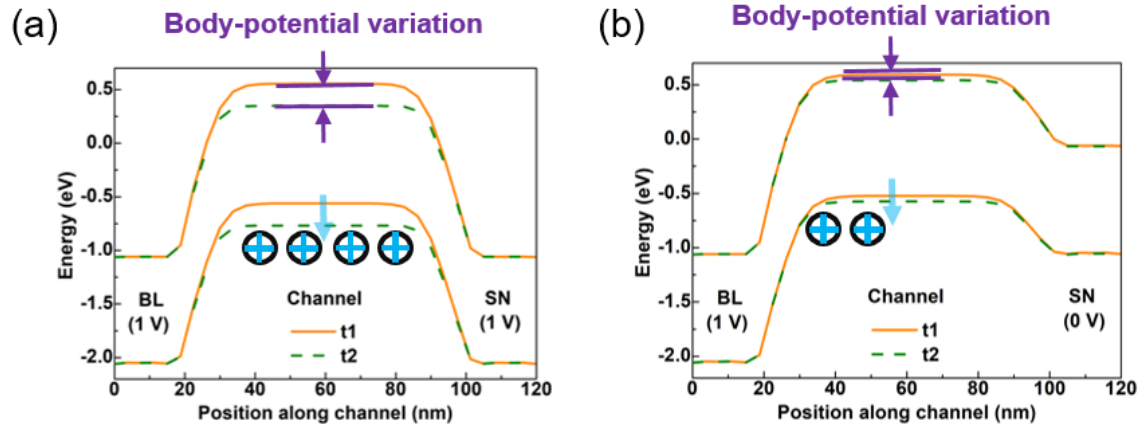


Figure 2.8. Energy-band profiles during the BL-high phase ($V_{BL} = 1V$) illustrating the degree of BPV in the vertical-channel DRAM cell. (a) SN = '1' state and (b) SN = '0'.

2.5 References

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Chapter 3. Parameter-Dependent Analysis of Floating-Body Effect

3.1 Impact of Gate Length on Floating-Body Effect

3.1.1 Introduction

The gate length (L_G) of a vertical-channel transistor (VCT) is a key parameter that determines the electrostatic coupling between the gate, drain, and floating body. As L_G is reduced, the drain electric field penetrates more deeply into the channel, increasing the junction field near both the BL and SN sides. This electrostatic weakening enhances band-to-band tunneling (BTBT)-induced GIDL and strengthens the floating body effect (FBE). To clarify this dependence, the electrical characteristics and internal field profiles were analyzed for $L_G = 30, 60, 90,$ and 120 nm.

Figure 3.1(a) shows the transfer characteristics for $L_G = 30$ – 120 nm. As the gate length decreases, the subthreshold current slightly increases, and the turn-on behavior shifts marginally, indicating weaker electrostatic gate control and stronger sensitivity to the drain field.

The physical origin of this behavior becomes clear when examining the electric-field distribution along the channel. Figure 3.1(b) presents the longitudinal electric field for the ‘1’ state. Short- L_G devices exhibit significantly higher peak junction fields at both the BL and SN sides, which directly enhances BTBT-induced GIDL and increases the hole generation rate inside the floating body.

Figure 3.1(c) shows the corresponding electric-field distribution for the ‘0’ state. In this bias condition, GIDL occurs only at the BL side, yet the same gate-length dependence persists: shorter devices maintain the strongest junction fields and therefore the greatest potential for FBE-induced disturbance.

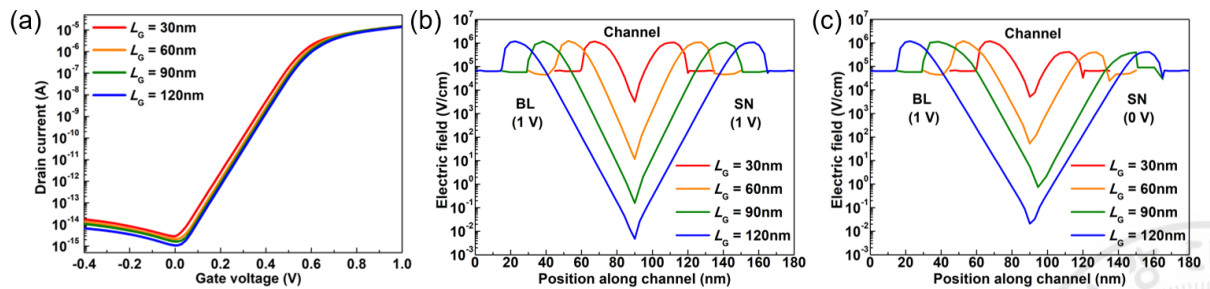


Figure 3.1. Impact of gate-length scaling on device electrostatics. (a) Transfer characteristics (I_D – V_G) for different gate lengths ($L_G = 30$ – 120 nm). (b) Longitudinal electric-field distribution in the ‘1’ state (BL = 1 V, SN = 1 V). (c) Longitudinal electric-field distribution in the ‘0’ state (BL = 1 V, SN = 0 V).

These results collectively indicate that the increased field penetration in Figure 3.1(b) and (c) is the underlying cause of the leakage and retention degradation observed later in the dynamic and static characteristics.

3.1.2 Dynamic Retention Characteristics

The dynamic-retention behavior directly reflects how the L_G -dependent drain-field intensifies the FBE. Figure 3.2(a) shows the transient V_{SN} response of the '1' state under BL-voltage toggling. Consistent with the electric-field trends in Figure 3.1(b), shorter- L_G devices exhibit a larger downward drift of V_{SN} because stronger BL- and SN-side GIDL generates more holes during each BL-high phase, raising the body potential and reducing the channel barrier.

Figure 3.2(b) presents the corresponding V_{SN} evolution for the '0' state. In this condition, GIDL originates only from the BL junction, yet the same L_G -dependent trend appears: devices with shorter channel lengths accumulate more GIDL-generated holes, resulting in a larger upward increase in V_{SN} over successive BL pulses.

Figure 3.2(c) quantifies the total V_{SN} fluctuation after ten BL toggles. In both storage states, the disturbance monotonically increases as L_G decreases, confirming that the enhanced junction electric field in short- L_G devices directly strengthens the floating body effect and accelerates dynamic-retention degradation. In addition to the enhanced junction fields shown in Figure 3.2(b) and (c), the increased subthreshold leakage of short- L_G devices observed in Figure 3.2(a) further accelerates the V_{SN} drift during dynamic operation.

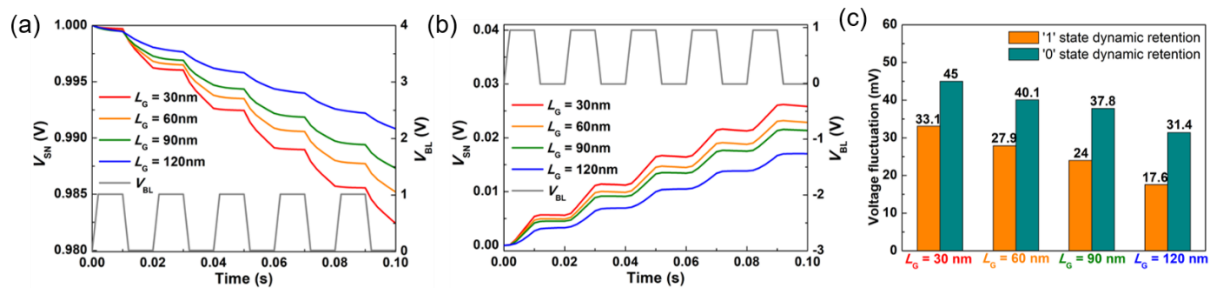
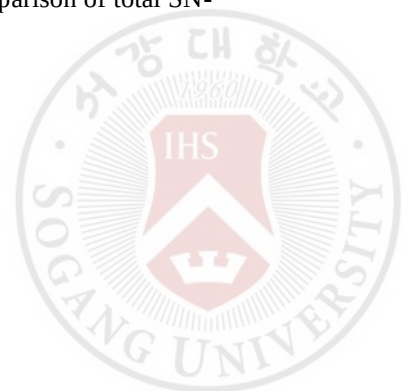


Figure 3.2. Dynamic-retention characteristics of the vertical-channel DRAM cell for different gate lengths ($L_G = 30, 60, 90, 120$ nm). (a) Transient SN-voltage response for the '1' state during BL-voltage toggling. (b) Transient SN-voltage response for the '0' state under the same operation condition. (c) Comparison of total SN-voltage fluctuation after ten BL-toggling cycles for both storage states.



3.1.3 Static-Retention Characteristics

Static retention was evaluated by maintaining the bit line at half the supply voltage while the word line remained off. Figure 3.3(a) shows the temporal evolution of V_{SN} when the cell stores a ‘1’. In this case, V_{SN} gradually decreases over time, and the reduction becomes more significant as L_G decreases. Short-channel devices experience stronger drain-induced electric fields, which continue to generate holes inside the floating body even without BL toggling. This sustained hole generation raises the body potential and lowers the channel barrier, accelerating the downward drift of V_{SN} .

Figure 3.3(b) presents the corresponding storage-node behavior when the cell holds a ‘0’. Here, the V_{SN} increases during the retention period because SN-to-body leakage becomes more pronounced as the body potential rises due to BL-side GIDL. As with the ‘1’ state, shorter gate lengths exhibit stronger junction fields, producing more holes and causing a steeper upward drift in V_{SN} . Longer channels, with weaker junction fields, show much smaller changes.

Figure 3.3(c) summarizes the total V_{SN} variation after one second for both storage states. The overall disturbance increases monotonically with decreasing L_G , confirming that static retention—similar to dynamic retention—is fundamentally limited by the enhanced floating body charging and increased off-state leakage characteristic of short-channel devices.

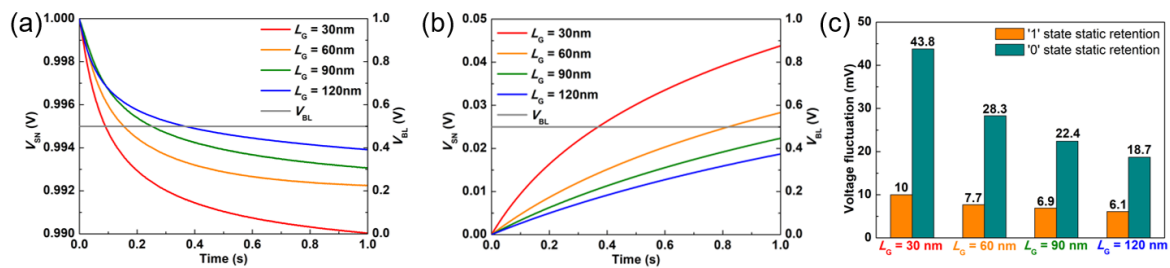
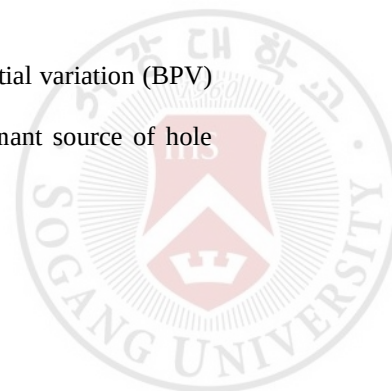


Figure 3.3. Static-retention characteristics of the vertical-channel DRAM cell for different gate lengths ($L_G = 30, 60, 90, 120$ nm). (a) Temporal V_{SN} for the ‘1’ state under $BL = V_{DD}/2$ and WL -off conditions. (b) Corresponding V_{SN} for the ‘0’ state. (c) Total V_{SN} fluctuation after 1 s of retention .

3.1.4 Evaluation of Pure Floating-Body Effect

To directly quantify the intrinsic strength of the FBE, the hole generation and body-potential variation (BPV) during the BL-high phase were analyzed for all gate lengths. Because GIDL is the dominant source of hole



injection into the floating-body, the spatial hole-density profile provides a clear indication of how strongly each device accumulates charge under high drain bias.

Figure 3.4(a) shows the hole-density distribution for the ‘1’ state when both BL and SN are biased at 1 V. Strong GIDL is generated at both junctions in this condition, and the amount of hole injection increases markedly as the gate length decreases. Short-channel devices exhibit deeper drain-field penetration and higher peak electric fields, resulting in significantly larger hole populations within the body region.

Figure 3.4(b) presents the corresponding hole-density profile for the ‘0’ state, where only the BL junction is biased at 1 V. Although hole injection occurs only from one side, the L_G dependence remains consistent. Shorter devices still produce noticeably higher hole density due to the intensified BL-side electric field, whereas longer devices show substantially reduced hole generation.

The accumulated holes raise the floating body potential, and the magnitude of this body-potential rise serves as a direct, leakage-independent indicator of FBE strength. Figure 3.4(c) summarizes the extracted BPV for the ‘1’ and ‘0’ states across the range of gate lengths. In both cases, BPV becomes larger as the gate length decreases, confirming that stronger drain-induced fields in short channels drive more pronounced floating body charging.

These results demonstrate that the severity of FBE is fundamentally governed by the amount of GIDL-induced hole generation during the BL-high phase. Shorter gate lengths amplify junction electric fields, increase hole injection, and ultimately produce larger BPV, directly explaining the stronger retention degradation observed in both dynamic and static conditions.

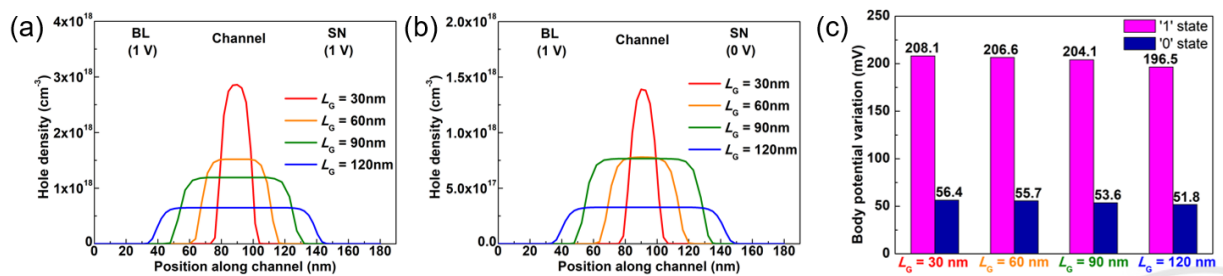


Figure 3.4. Evaluation of the pure FBE under the BL-high condition for different gate lengths ($L_G = 30, 60, 90, 120$ nm). (a) Hole-density distribution along the channel for the ‘1’ state (BL = 1 V, SN = 1 V) at different gate lengths. (b) Hole-density distribution for the ‘0’ state (BL = 1 V, SN = 0 V). (c) Extracted BPV during the BL-high phase.

3.2 Impact of Body Thickness on Floating-Body Effect

3.2.1 Introduction

In addition to gate length, the body thickness (T_{Body}) critically influences the strength of the FBE in vertical-channel transistors. Thinner bodies strengthen gate–channel coupling, suppressing floating body potential fluctuations and mitigating FBE-induced retention degradation [3.2].

However, when T_{Body} is reduced to a few nanometers, quantum confinement effects (QCE) arise, as illustrated in Figure 3.5. QCE occurs when the channel dimension approaches the de Broglie wavelength, restricting motion in one or more directions [3.3]. This spatial confinement quantizes the subbands in the conduction and valence bands, effectively widening the bandgap and shifting the inversion-charge centroid away from the interface. As a result, the carrier distribution, subband structure, and threshold voltage are altered, and the available inversion charge—and thus drive current—can be reduced [3.4]–[3.6].

These quantum effects can counteract the electrostatic benefit of FBE suppression, introducing a trade-off between improved gate control and degraded current capability. To examine this trade-off, TCAD simulations were performed while varying T_{Body} from 6 to 3 nm under identical gate, overlap, and doping conditions. We first analyze the intrinsic electrostatic effect by excluding QCE, and then include QCE to evaluate its influence on floating body behavior and overall current characteristics.

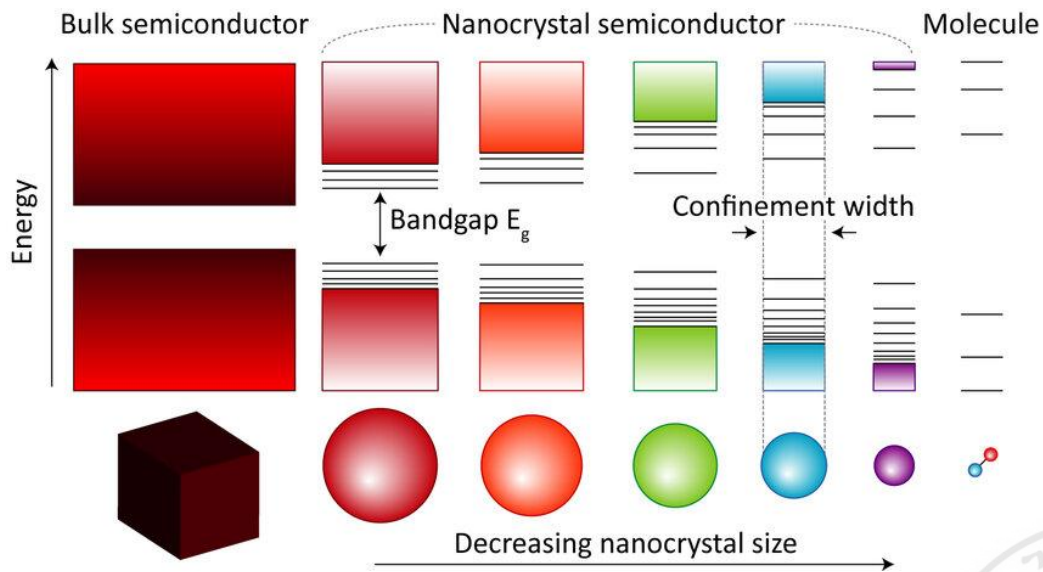


Figure 3.5. Schematic illustration of quantum confinement and its effect on electronic states as size approaches the nanometer regime [3.3].



3.2.2 Electrostatic Effect of Body-Thinning (without QCE)

(1) Dynamic-Retention Characteristics

Figure 3.6 presents the dynamic-retention characteristics for $T_{\text{Body}} = 3, 4, 5, \text{ and } 6 \text{ nm}$. As shown in Figure 3.6(a) and 3.6(b), thinning the body reduces the fluctuation of V_{SN} in both the '1' and '0' states. This improvement originates from the suppressed FBE: a thinner body provides stronger gate-to-body electrostatic control and reduces the volume available for hole accumulation during the BL-high phase, thereby mitigating GIDL-driven charging.

In addition to the FBE-related suppression, thinner channels also exhibit lower off-state leakage due to enhanced channel depletion, which further slows the drift of V_{SN} during repeated BL toggling. As a combined result of both the reduced hole generation and the decreased off-leakage, the total V_{SN} variation after ten toggles, summarized in Figure 3.6(c), decreases monotonically as T_{Body} becomes thinner.

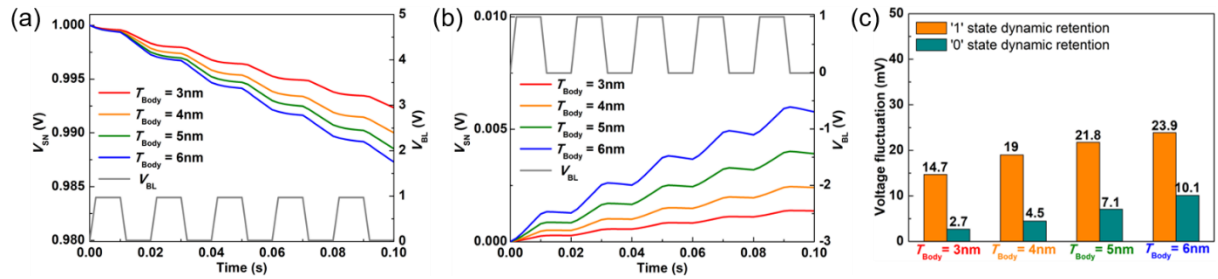
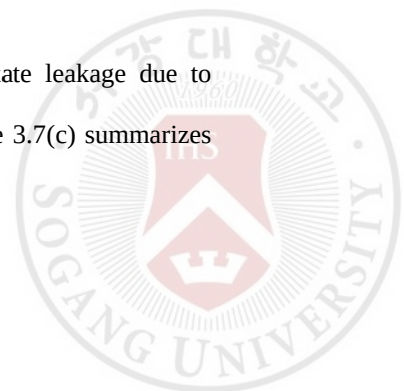


Figure 3.6. Dynamic retention for different T_{Body} . (a) Transient V_{SN} for the '1' state during BL toggling. (b) Transient V_{SN} for the '0' state. (c) Total V_{SN} variation after ten toggles for both states, showing reduced fluctuation with thinner T_{Body} .

(2) Static-Retention Characteristics

Figure 3.7 shows the static-retention behavior with the word line turned off and the bit line held at $V_{\text{DD}}/2$. Figure 3.7(a) and Figure 3.7(b) illustrate that the drift of V_{SN} becomes noticeably smaller as T_{Body} decreases for both the '1' and '0' states. This improvement results from the reduced FBE in thinner channels: a thinner body has stronger gate-to-body electrostatic control and a smaller neutral-body volume, which limits hole accumulation and stabilizes the body potential during the hold period.

In addition to this FBE-related suppression, thinner devices also exhibit lower off-state leakage due to enhanced channel depletion, further reducing the slow drift of V_{SN} under static bias. Figure 3.7(c) summarizes



the total voltage change after one second of retention and confirms that static-retention stability improves monotonically as T_{Body} becomes thinner.

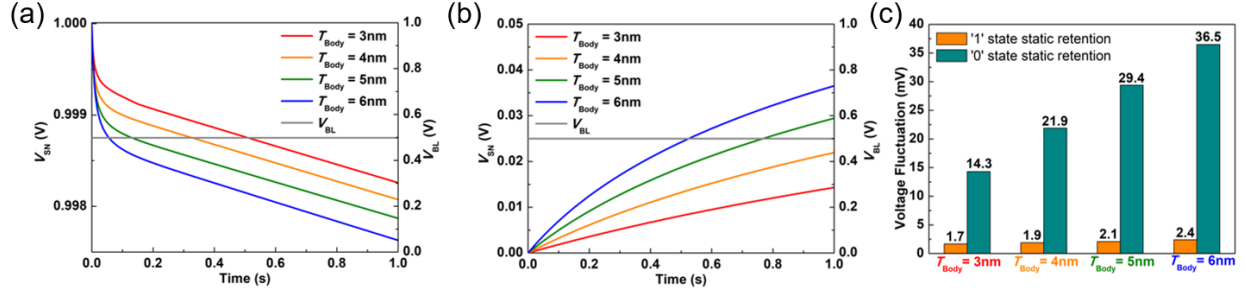


Figure 3.7. Static retention for different T_{Body} . (a) Temporal V_{SN} for the '1' state at $BL = V_{\text{DD}}/2$, WL -off. (b) Corresponding result for the '0' state. (c) Total V_{SN} variation after 1 s of retention.

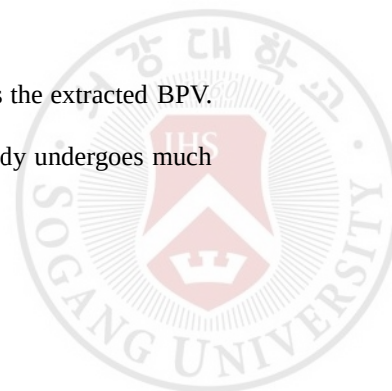
(3) Evaluation of Pure Floating-Body Effect

To evaluate how body-thickness scaling influences the intrinsic FBE, the hole-density distribution and the resulting BPV during the BL -high phase were analyzed for various T_{Body} . Since GIDL-generated holes are the primary source of floating body charging, their spatial distribution directly reflects the strength of the FBE.

Figure 3.8(a) shows the hole-density distribution in the '1' state, where both the BL and SN are biased at 1 V. As the body becomes thinner, the number of generated holes decreases noticeably. A thinner body experiences stronger gate-to-body electrostatic control, which reduces the junction electric field and suppresses BTBT-induced hole generation. Furthermore, the neutral-body volume available to store generated holes becomes smaller as T_{Body} decreases.

Figure 3.8(b) presents the hole-density distribution for the '0' state, with only the BL junction under high bias. Despite the asymmetric biasing, the scaling trend is the same: thinner bodies accumulate fewer holes due to the reduced body volume and diminished BL -side field strength. This consistent behavior across the two data-retention states indicates that electrostatic confinement—a direct consequence of thinner body thickness—is the dominant mechanism for reducing hole generation.

The effect of this reduced hole population is reflected in Figure 3.8(c), which summarizes the extracted BPV. The BPV decreases monotonically as T_{Body} becomes thinner, indicating that the floating body undergoes much



weaker potential modulation under BL-high conditions. This trend aligns with the hole-density analysis, confirming that thinning the body effectively suppresses FBE by reducing both the generation and storage of holes.

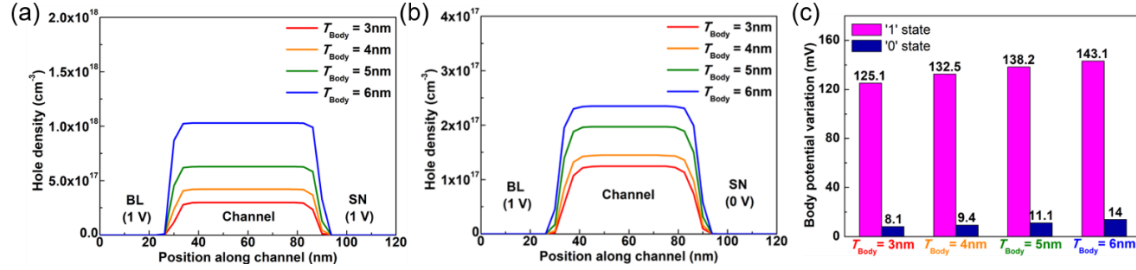


Figure 3.8. Evaluation of the pure FBE for different body thicknesses. (a) Hole-density distribution along the channel in the '1' state (BL = 1 V, SN = 1 V). (b) Hole-density distribution in the '0' state (BL = 1 V, SN = 0 V). (c) Extracted BPV during the BL-high phase for $T_{\text{Body}} = 3, 4, 5,$ and 6 nm.

3.2.3 Influence of Quantum Confinement Effect (QCE)

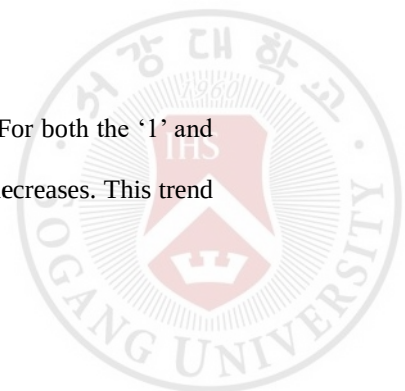
When QCE is included, band-edge quantization modifies the charge distribution and thereby alters retention behavior [3.4]-[3.6]. The upward shift in the conduction-band edge and the reduced inversion charge suppress hole generation near the drain, further reducing FBE.

(1) Dynamic-Retention Behavior and Body-potential Variation under QCE

Figure 3.9 presents the dynamic-retention characteristics with QCE enabled for various body thicknesses. The temporal evolution of V_{SN} in Figure 3.9(a) ('1' state) and Figure 3.9(b) ('0' state) shows that the fluctuation in V_{SN} is reduced across all T_{Body} values compared to the QCE-excluded case. This improvement demonstrates that quantum confinement enhances gate-to-body control and suppresses floating body charging during the BL-high phase.

The total V_{SN} fluctuation after ten toggles, summarized in Figure 3.9(c), further confirms that QCE mitigates retention disturbance for both states. The benefit is more pronounced in thinner bodies, where QCE increases the effective bandgap and strengthens depletion, leading to a substantial reduction in both GIDL-induced hole generation and off-state leakage.

The extracted BPV shown in Figure 3.9(d) provides direct evidence of this suppression. For both the '1' and '0' states, BPV decreases when QCE is enabled, and the reduction becomes larger as T_{Body} decreases. This trend



indicates that quantum confinement reduces the effective density of states and strengthens channel depletion, which limits the amount of GIDL-generated holes that can accumulate in the floating body. As a result, BPV during the BL-high phase becomes weaker.

Collectively, these results confirm that QCE acts as an additional electrostatic mechanism that suppresses FBE. Thinner bodies benefit the most because quantum confinement is stronger in narrow geometries, yielding a combined effect of reduced hole generation, lower BPV, and improved dynamic-retention stability.

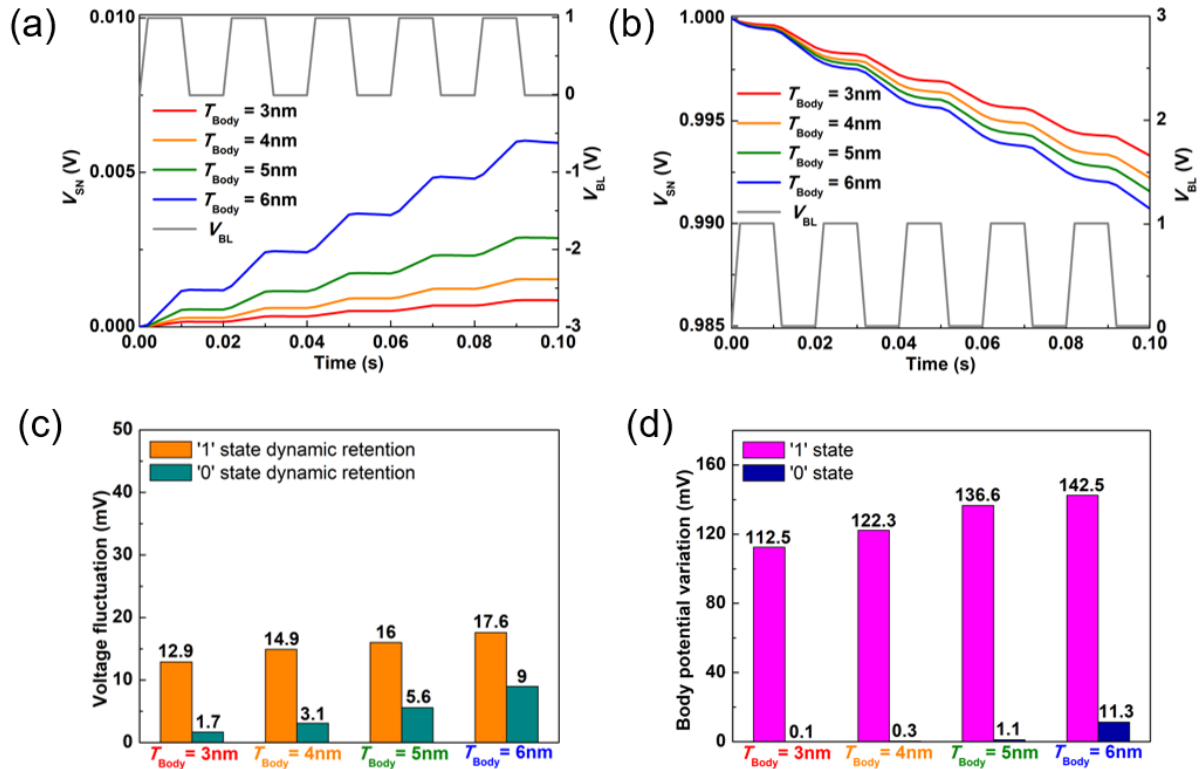
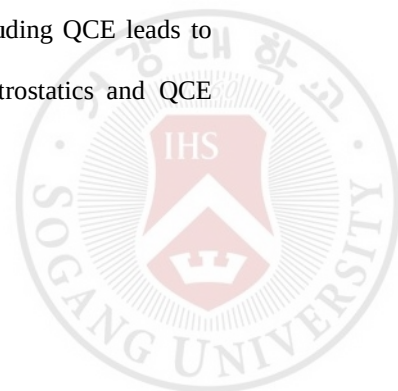


Figure 3.9. Dynamic-retention characteristics with QCE included for $T_{\text{Body}} = 3, 4, 5,$ and 6 nm. (a) Transient V_{SN} for the '1' state. (b) Transient V_{SN} for the '0' state. (c) Total V_{SN} variation after ten BL-toggling cycles. (d) Extracted BPV during the BL-high phase, showing reduced floating body modulation with QCE.

(2) Static-Retention Behavior under QCE

Figure 3.10 shows the static retention with QCE included. Figures 3.10(a) and 3.10(b) show smaller V_{SN} drift for thinner devices, consistent with the dynamic trends. Figure 3.10(c) confirms that including QCE leads to further improvements in retention relative to the purely electrostatic case. Overall, electrostatics and QCE together suppress floating body potential modulation in ultra-thin channels.



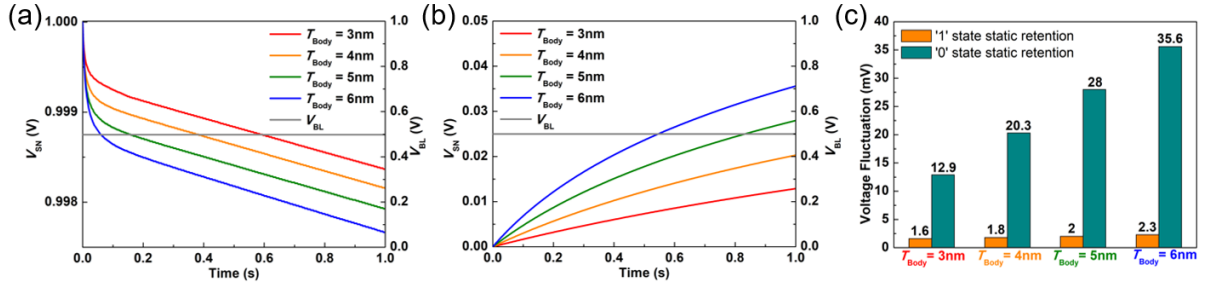


Figure 3.10. Static retention with QCE. (a) Temporal V_{SN} for the '1' state at $BL = V_{DD}/2$, WL -off. (b) Corresponding result for the '0' state. (c) Total V_{SN} variation after 1 s.

3.2.4 Trade-Off between FBE Suppression and Drive Current

Quantum confinement helps retention by suppressing the FBE, but it also alters the drive capability of the access device. To quantify this trade-off, we evaluate how $T_{Body} = 3\text{--}6\text{ nm}$ affects the transfer characteristics and normalized figures of merit that isolate the impact of QCE.

Figure 3.11(a) shows simulated I_D - V_G characteristics without QCE. As T_{Body} decreases, the subthreshold branch shifts only slightly while the strong-inversion current remains high, indicating that—absent quantization—aggressive thinning is largely beneficial due to strengthened electrostatic control. With QCE enabled, as shown in Figure 3.11(b), thinning has a much stronger impact: for $T_{Body} = 3\text{ nm}$ the turn-on shifts to higher V_G and the attainable on-current is reduced [3.4]–[3.6]. This degradation stems from carrier quantization in the ultra-thin body, which pushes the conduction-band edge upward and suppresses inversion charge; consequently, more gate bias is required to reach the same current level, and even the high- V_G current is capped. The difference weakens for thicker bodies, where quantization is less severe and the I_D - V_G curve approaches the no-QCE case.

To more directly compare the competing impacts of quantum confinement on current drivability and FBE suppression, Figures 3.11(c) and 3.11(d) separately plot the QCE-enabled and QCE-disabled characteristics instead of using normalized ratios as in the earlier version of this analysis.

Figure 3.11(c) shows that both the BPV and dynamic-retention V_{SN} fluctuation steadily decrease as T_{Body} is reduced, reflecting the natural suppression of the FBE in thinner bodies. Importantly, the QCE-applied curves exhibit consistently lower BPV values than the non-QCE curves across all thicknesses, confirming that quantum confinement further suppresses hole accumulation and stabilizes the floating body. A similar trend is observed in

the dynamic-retention behavior in Figure 3.11(d), where QCE again yields smaller V_{SN} excursions compared to the non-QCE case, and both quantities improve monotonically with reduced T_{Body} .

However, while FBE-related metrics benefit from aggressive thinning, the on-current does not scale favorably. As shown in both figures, the on-current reduction induced by QCE becomes substantial below $T_{Body} \approx 5$ nm. This degradation arises because stronger confinement elevates the conduction-band edge and limits carrier injection, causing a steep decline in I_{On} when the body becomes thinner than ~ 5 nm.

Taken together, these results indicate that although the smallest body thicknesses provide the strongest suppression of FBE—especially when combined with QCE—the accompanying loss of drive current becomes increasingly detrimental below 5 nm. Accordingly, $T_{Body} \approx 5$ nm emerges as a practical scaling limit that provides an efficient balance: sufficient QCE-assisted stabilization of the floating body without incurring excessive on-current degradation.

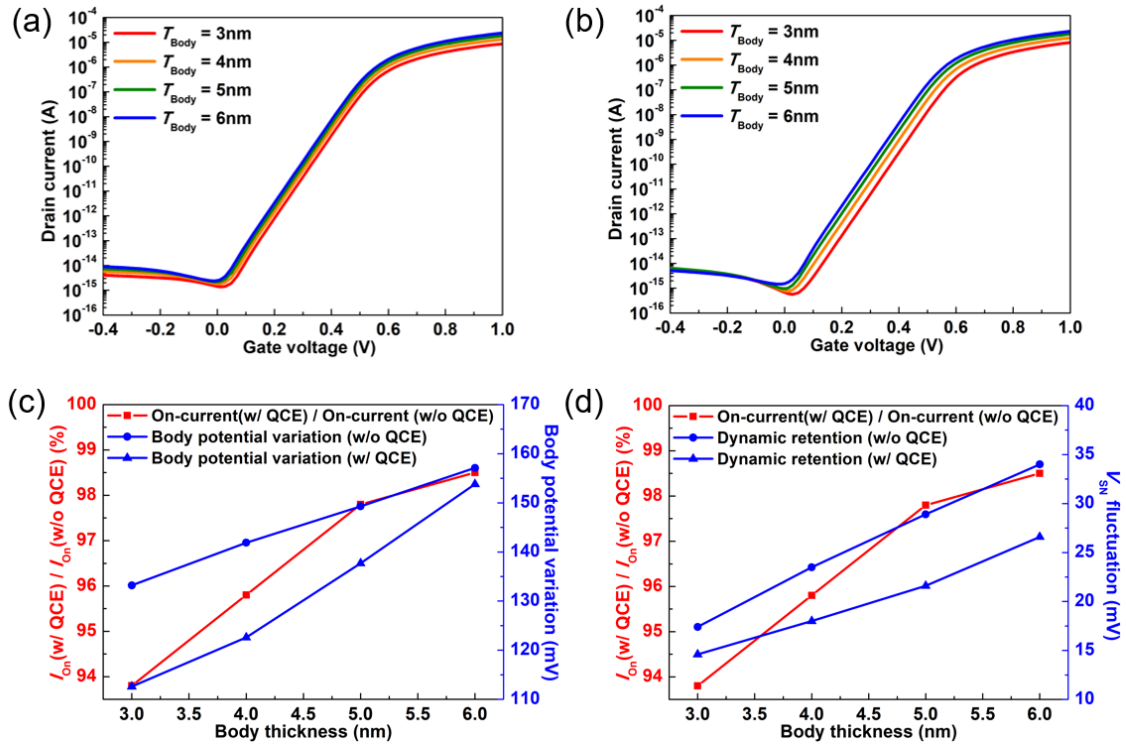
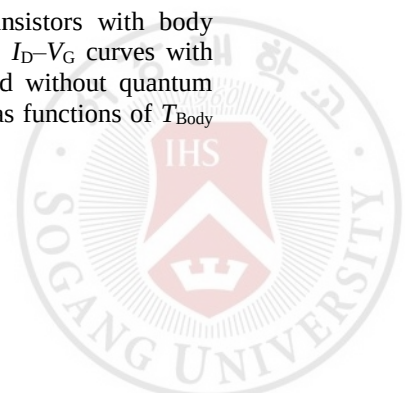


Figure 3.11. (a) Simulated I_D – V_G characteristics of vertical-channel DRAM access transistors with body thickness $T_{Body} = 3$ – 6 nm without quantum confinement effect (QCE). (b) Corresponding I_D – V_G curves with QCE included. (c) BPV and on-current characteristics versus T_{Body} for devices with and without quantum confinement effects (QCE). (d) Dynamic-retention V_{SN} fluctuation (ΔV_{SN}) and on-current as functions of T_{Body} under QCE-enabled and QCE-disabled conditions.



3.3 Impact of Drain(SN) Doping Concentration on Floating-Body Effect

3.3.1 Introduction

Engineering the drain electric field through lightly doped drain (LDD) structures has been widely employed to mitigate gate-induced drain leakage (GIDL) and associated floating-body effects (FBE) in scaled transistors [3.7]-[3.9]. The LDD region effectively reduces the local electric field at the drain junction, thereby suppressing band-to-band tunneling and minimizing hole generation in the floating-body.

In vertical-channel DRAM cells, the same concept can be applied by adjusting the storage-node (SN) side doping concentration, which corresponds to the drain during the ‘1’-state operation and the source during the ‘0’-state operation. Thus, controlling the SN-side doping concentration (N_D) serves as a practical means to emulate LDD behavior and analyze its influence on the FBE under realistic DRAM bias conditions.

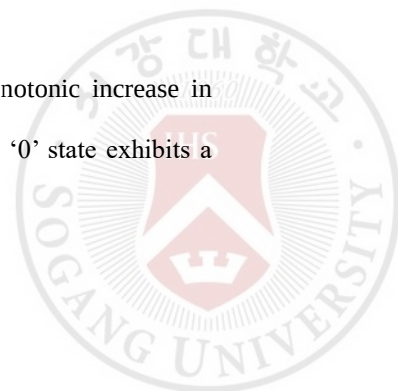
To systematically evaluate this effect, TCAD simulations were performed for $N_D = 1 \times 10^{19}$, 5×10^{19} , and $1 \times 10^{20} \text{ cm}^{-3}$ under fixed gate length, body thickness, and overlap geometry. The impact of SN-side doping concentration on dynamic retention, static retention, and BPV was analyzed, and the underlying mechanism was examined through the I_D - V_G characteristics in both forward and reverse operation modes.

3.3.2 Dynamic Retention Characteristics

Figure 3.12 presents the dynamic-retention characteristics as a function of SN-side drain doping concentration. When the SN side serves as the drain in the ‘1’ state, shown in Figures 3.12(a), higher N_D intensifies the drain electric field and enhances GIDL at the SN junction. This increases the amount of BTBT-generated holes injected into the floating body, resulting in larger V_{SN} fluctuations during BL toggling.

In contrast, the impact of SN-side doping on the ‘0’ state, shown in Figure 3.12(b), is more moderate. The BL-side GIDL strength remains nearly unchanged because the BL side acts as the drain, and its doping concentration is fixed. However, increasing SN doping still modifies the source-side junction field and the associated leakage paths, leading to noticeable but weaker changes in the body-potential dynamics compared to the ‘1’ state.

Figure 3.12(c) summarizes the total V_{SN} variation after ten BL-toggling cycles. A monotonic increase in fluctuation is observed for the ‘1’ state due to drain-side GIDL enhancement, whereas the ‘0’ state exhibits a



more gradual dependence on SN-side doping because the dominant BL-side field is unaffected by the doping split.

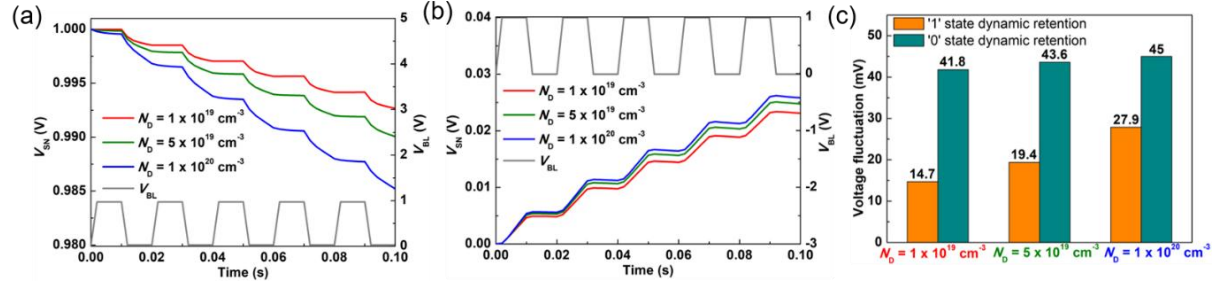


Figure 3.12. Dynamic retention for different drain-side doping concentrations ($N_D = 1 \times 10^{19}$, 5×10^{19} , and $1 \times 10^{20} \text{ cm}^{-3}$). (a) Transient V_{SN} for the '1' state. (b) Transient V_{SN} for the '0' state. (c) Total V_{SN} fluctuation after ten toggles.

3.3.3 Static Retention Characteristics

Figure 3.13 summarizes static-retention behavior for different dopings. In the '1' state, the SN region behaves as the drain; higher N_D produces a stronger field, resulting in faster V_{SN} decay due to intensified GIDL-hole generation as shown in Figure 3.13(a). In the '0' state, the roles reverse, and the SN side acts as the source. Under this condition, lowering N_D increases the effective source resistance and reduces electron injection, suppressing leakage and stabilizing V_{SN} as shown in Figure 3.13(b). Consequently, smaller doping mitigates FBE-induced drift but can also slow subsequent write/recovery. Figure 3.13(c) shows that lower SN-side doping consistently improves retention stability by reducing GIDL-related leakage and hole accumulation.

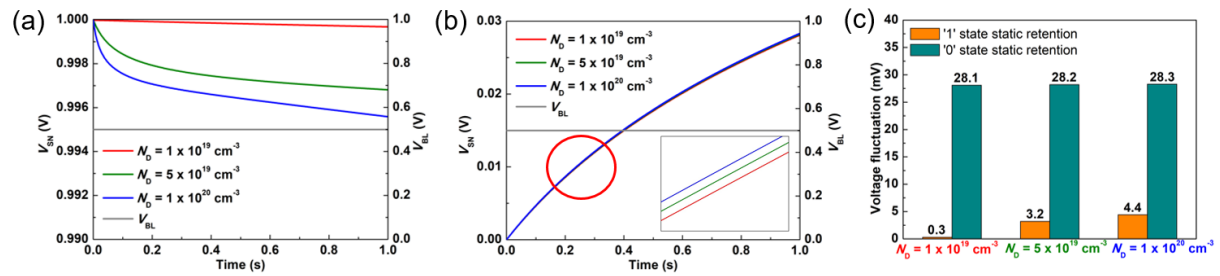


Figure 3.13. Static retention for different drain-side doping concentrations. (a) V_{SN} for the '1' state at $V_{BL} = V_{DD}/2$. (b) V_{SN} for the '0' state. (c) Total V_{SN} fluctuation after 1 s of retention.



3.3.4 Evaluation of Pure Floating-Body Effect

Figure 3.14 evaluates the pure FBE by examining the junction electric-field distribution and the resulting BPV for different SN-side doping concentrations. Figures 3.14(a) and 3.14(b) present the electric-field profiles for the ‘1’ and ‘0’ states, respectively.

In the ‘1’ state, the SN side serves as the drain, and its doping concentration directly determines the drain-junction electric field. As N_D increases, the SN-side electric field becomes significantly stronger, which enhances GIDL and increases the number of BTBT-generated holes injected into the floating body. Consequently, the BPV increases with higher N_D as shown in Figure 3.14(c). This confirms that strong SN-side fields intensify the FBE, whereas lowering N_D suppresses GIDL and stabilizes the body potential.

In the ‘0’ state, the BL side acts as the drain, and its doping concentration is fixed; therefore, the dominant BL-side GIDL remains unchanged across the doping split. However, modifying the SN-side doping still influences the floating body dynamics through changes in the source-side junction field. When N_D is low, the SN-side junction field becomes weak, making it easier for holes to discharge from the floating body toward the SN junction rather than being accumulated in the body. As a result, the floating body stores fewer holes during the BL-high phase, leading to a smaller BPV. As N_D increases, the stronger SN-side field facilitates greater charge redistribution within the floating body during the BL-high phase, resulting in an increased BPV, as captured in Figure 3.14(c).

Overall, these results demonstrate that SN-side doping plays an important role in shaping the FBE. Higher doping enhances the junction field and strengthens FBE-induced BPV, while lower doping weakens the junction field and suppresses FBE. Thus, SN-side junction engineering provides an effective knob for controlling FBE behavior in vertical-channel DRAM cells.

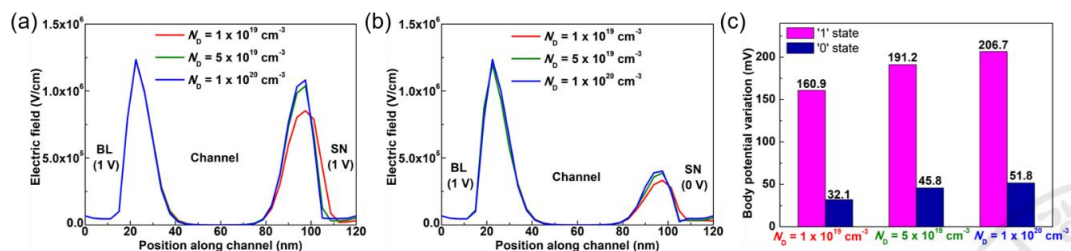


Figure 3.14. Impact of SN-side doping on the FBE. (a) ‘1’ state electric-field profiles for different N_D . (b) ‘0’ state electric-field profiles. (c) Extracted BPV during the BL-high phase showing stronger body-potential modulation with higher N_D .

3.3.5 Mechanistic Analysis via Transfer Characteristics

Figure 3.15 examines I_D - V_G curves under forward and reverse operation. In Figure 3.15(a) (forward mode, ‘1’ state), lowering N_D reduces the drain field, suppressing BTBT and GIDL; thus, both off-state leakage and body-charging rate decrease, leading to weaker FBE and improved retention. In Figure 3.15(b) (reverse mode, ‘0’ state), reducing the SN-side doping effectively increases source resistance and the injection barrier, decreasing on-current. This implies that although FBE is only moderately alleviated by lower N_D write/read speeds can be degraded because carrier injection from the SN side becomes less efficient.

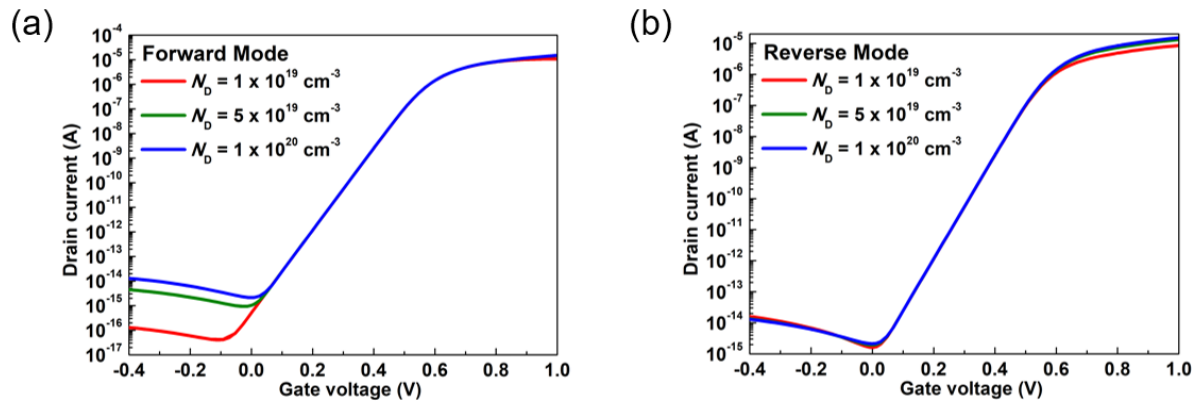


Figure 3.15. I_D - V_G for different SN-side doping concentrations. (a) Forward mode (‘1’ state, SN as drain). (b) Reverse mode (‘0’ state, SN as source).

3.3.6 Discussion and Summary

The SN-side doping concentration plays a decisive role in balancing floating body suppression and access performance in vertical-channel DRAM cells. As demonstrated by the simulation results, lowering the SN-side N_D effectively relaxes the drain electric field when the SN serves as the drain during the ‘1’-state operation. This reduction in local field strength suppresses GIDL, thereby minimizing hole generation and accumulation in the floating body. Consequently, devices with moderately reduced N_D exhibit smaller BPV and improved retention stability, confirming that electric-field moderation through doping control is an effective method to alleviate the FBE.

However, this improvement comes at the expense of current drivability. When the SN region operates as the source during the ‘0’-state operation, a lower doping concentration increases the series resistance and the effective injection barrier, leading to degraded carrier transport and slower data access. This trade-off becomes particularly pronounced at very low N_D , where the weakened source injection severely limits the on-current and

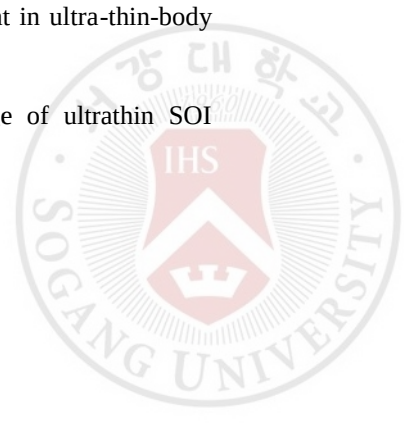
elongates the write and restore times. Therefore, although a reduction in SN-side doping clearly contributes to improved FBE stability, it cannot serve as an isolated or comprehensive solution for high-speed DRAM operation.

The combined analysis of dynamic and static retention behaviors reveals that optimal device performance requires a balance between electrostatic field control and charge transport efficiency. Excessive doping reduction overemphasizes FBE mitigation at the cost of performance, whereas high doping restores speed but reintroduces strong GIDL and floating body charging. This dual constraint underscores the importance of local field engineering—rather than simple doping modification—as the most practical and scalable strategy. Techniques such as graded junctions, optimized overlap geometries, and asymmetric drain extensions can locally suppress GIDL and stabilize the floating body while maintaining sufficient current drive.

In summary, tuning the SN-side doping concentration provides valuable insight into the physical mechanisms governing the FBE, but the results clearly indicate that robust FBE mitigation must rely on electrostatic field optimization through structural and material engineering. This conclusion naturally motivates the exploration of gate-stack and field-profile engineering approaches—such as split-gate and multi-work-function designs—discussed in Chapter 4.

3.4 References

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Chapter 4. Mitigation Strategies of Floating-Body Effect

4.1 Split-Gate Structure for Floating-Body Effect Mitigation

Chapter 3 showed that the FBE can be mitigated by (i) reducing the body thickness to strengthen electrostatic control or (ii) relaxing the drain-side electric field to suppress gate-induced drain leakage (GIDL) [4.1], [4.2]. While T_{Body} naturally decreases with technology scaling up to the regime where quantum confinement effects (QCE) become significant, a practical and layout-compatible route is to engineer the local junction field and thereby reduce GIDL [4.3].

Building on this idea, prior work proposed a multi-WL (split-gate) configuration for vertical DRAM access transistors, in which an auxiliary gate segment adjacent to the state-dependent drain junction is biased at a positive voltage to reduce the gate–drain field and suppress GIDL-induced hole generation in the floating body. [4.4]. Figure 4.1 illustrates the operating principle.

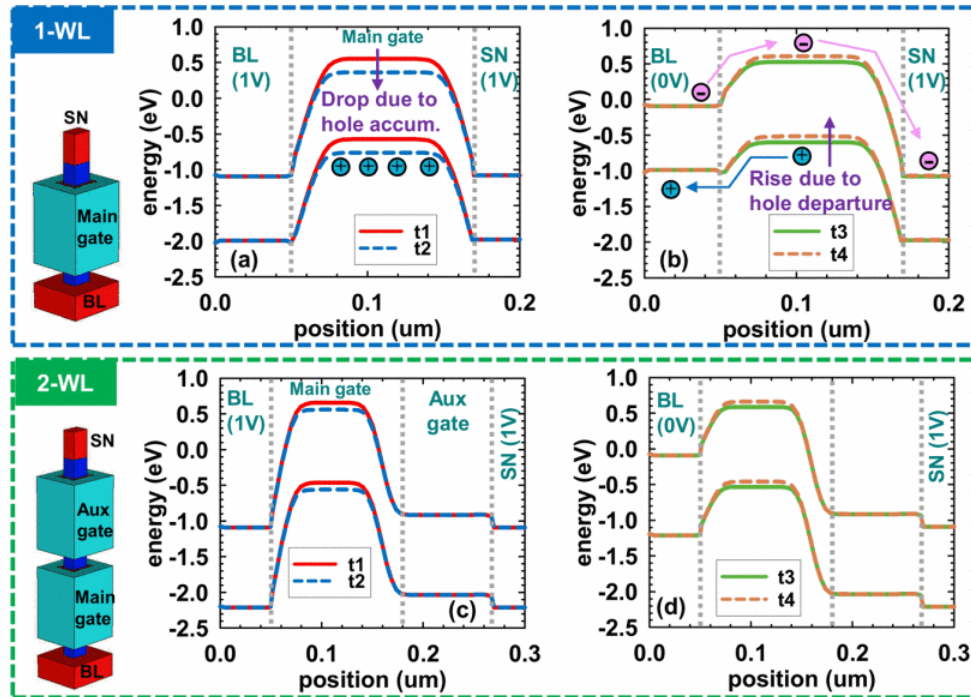


Figure 4.1. Energy-band diagrams of single-WL and dual-WL (split-gate) VCTs. (a,b) single-WL device under BL-high and BL-low, respectively. (c,d) dual-WL device under BL-high and BL-low, respectively.

As shown in Figures 4.1(a) and 4.1(b), the conventional single-WL device experiences strong GIDL at the drain-side junction under a high BL voltage (BL = 1 V), generating holes in the floating body. This increases the

body potential, which appears as a downward shift of the energy-band profile in the band diagram. When the BL returns to 0 V, these holes discharge primarily toward the BL, and the body-potential decreases (corresponding to an upward shift in the band diagram), resulting in cycle-to-cycle variation of the body voltage.

As illustrated in Figures 4.1(c) and 4.1(d), the auxiliary gate in the dual-WL (split-gate) structure is positively biased during BL toggling. This reduces the effective gate–drain field at the active drain junction, suppresses hole generation, moderates the body-potential excursion, and improves stability under dynamic operation.

From an implementation perspective, however, the multi-WL approach is a prior-art mitigation with notable overheads: it increases (i) array-level complexity (extra routing and timing control for the auxiliary word line) and (ii) process complexity (segmented-gate formation and alignment), making large-scale DRAM integration more challenging compared to a conventional single-WL VCT.

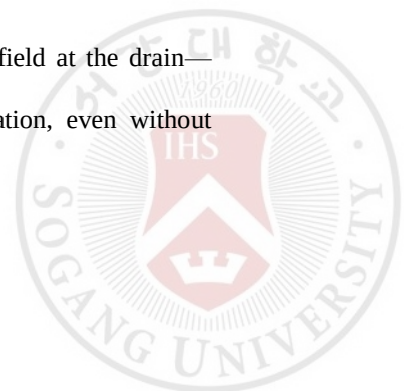
4.2 Dual Work-Function Gate Engineering for Floating-Body Effect Mitigation

4.2.1 Introduction

To suppress GIDL while maintaining a single WL architecture, this work introduces a dual work-function (dual-WF) gate structure in which two vertically stacked gate metals—Metal 1 on the BL side and Metal 2 on the SN side—are assigned different work functions. As illustrated in Figure 4.2(a), this configuration preserves the standard single-WL topology while enabling localized electrostatic control through the engineered work-function split.

The impact of this dual-WF arrangement on the energy-band profile is shown in Figure 4.2(b). A higher work function on the source-side gate (Metal 1) raises the source–channel barrier, suppressing electron leakage from the source into the channel. Conversely, assigning a lower work function to the drain-side gate (Metal 2) relaxes the drain electric field under BL-high bias, thereby reducing GIDL and limiting the amount of hole generation at the drain junction.

This combined modulation—raising the barrier at the source and relaxing the electric field at the drain—reduces unwanted leakage paths and improves retention stability during dynamic operation, even without altering the WL configuration.



By coordinating the work-function levels of the two gate segments, the dual-WF gate achieves simultaneous leakage reduction and enhanced retention characteristics, while retaining the simplicity and scalability of a single-word-line structure.

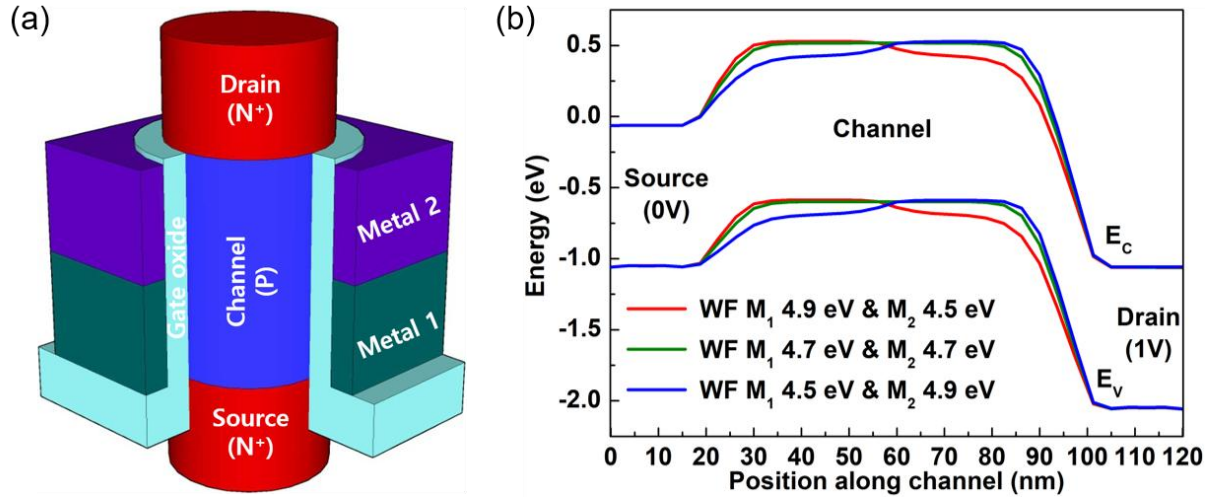


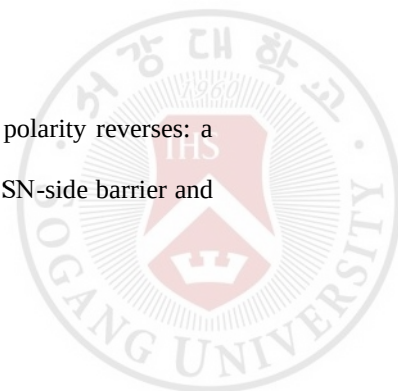
Figure 4.2. Dual-WF vertical-channel transistor structure and work-function-dependent energy-band modulation. (a) 3D schematic of the dual-WF gate consisting of Metal 1 (BL side) and Metal 2 (SN side). (b) Energy-band profiles for different WF pairs ($M_1/M_2 = 4.9/4.5$, $4.7/4.7$, and $4.5/4.9$ eV) under the '1'-state bias condition.

4.2.2 Dynamic Retention Characteristics

The impact of dual work-function (dual-WF) engineering on dynamic-retention behavior was evaluated for three WF combinations of the stacked gate metals (M_1 on the BL side, M_2 on the SN side): $4.9/4.5$ eV, $4.7/4.7$ eV, and $4.5/4.9$ eV. Figure 4.3 shows the corresponding transient V_{SN} responses for the '1' and '0' states under repeated BL toggling, and the total fluctuation after ten cycles. Figure 4.4 presents the associated energy-band profiles under BL-low ('1' state) and BL-high ('0' state) bias conditions.

In the '1' state, the SN side acts as the drain. A lower work function for M_2 ($4.9/4.5$ eV) reduces the drain-side electric field, as seen in the flattened band slope in Figure 4.4(a). This suppressed field weakens GIDL-induced hole generation, resulting in the smallest V_{SN} drop among the examined WF pairs in Figure 4.3(a) and 4.3(c). Conversely, when M_2 has the higher work function ($4.5/4.9$ eV), the stronger drain-side field enhances GIDL, producing the largest V_{SN} degradation for the '1' state.

In the '0' state, the effective drain shifts to the BL side. Accordingly, the optimal WF polarity reverses: a higher work function for M_2 and a lower work function for M_1 ($4.5/4.9$ eV) strengthens the SN-side barrier and



suppresses GIDL at the BL junction under BL-high, reducing the V_{SN} rise observed in Figure 4.3(b) and 4.3(c). The opposite polarity (4.9/4.5 eV) weakens the barrier and increases hole injection toward the body, yielding the poorest retention in the '0' state.

These results highlight an inherent state-dependent trade-off: the WF polarity that stabilizes the '1' state inevitably degrades the '0' state, and vice versa, because the drain-side junction—and therefore the location requiring field relaxation—changes with the stored state. Thus, dual-WF gate engineering provides a tunable mechanism to modulate junction fields and shape dynamic-retention behavior, but its effectiveness is fundamentally bounded by the opposing WF requirements of the '1' and '0' states.

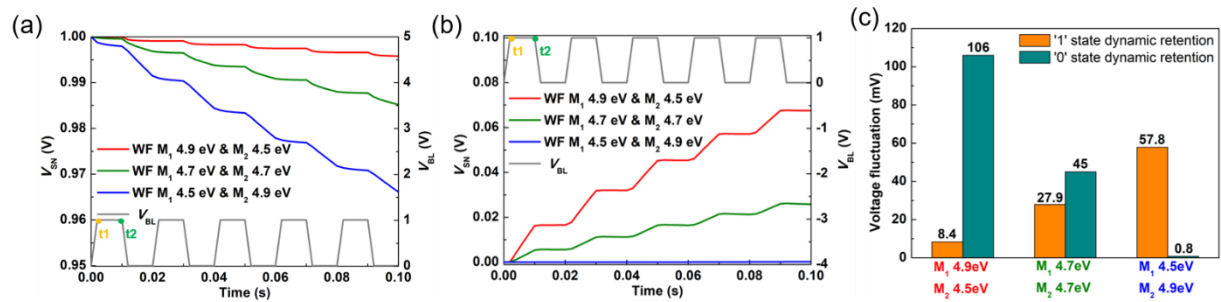


Figure 4.3. Dynamic-retention characteristics for various WF pairs ($M_1/M_2 = 4.9/4.5, 4.7/4.7, 4.5/4.9$ eV). (a) SN-voltage response for the '1' state. (b) for the '0' state. (c) Total SN-voltage fluctuation after ten BL-toggling cycles.

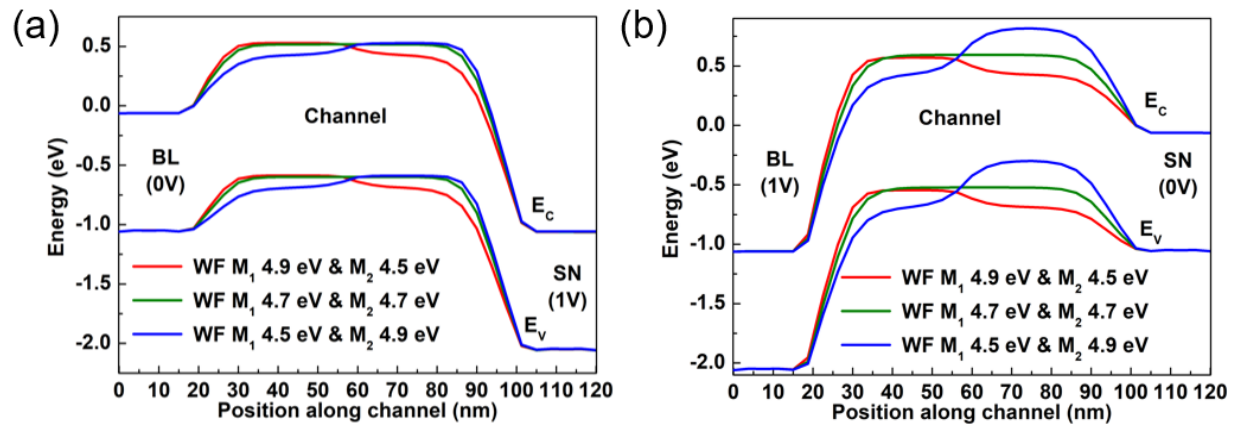


Figure 4.4. Energy-band diagrams of the dual-WF vertical transistor under dynamic-retention bias. (a) Band profile for the '1' state (BL = 0 V, SN = 1 V). (b) Band profile for the '0' state (BL = 1 V, SN = 0 V).



4.2.3 Static Retention Characteristics

The impact of the dual-WF gate on static retention was evaluated under WL-off and $BL = V_{DD}/2$ bias. Unlike dynamic retention, where transient GIDL-driven hole accumulation modulates the floating body potential, static retention is governed mainly by steady-state junction leakage. Therefore, the effect of the dual-WF configuration appears through the way it reshapes the BL- and SN-side junction barriers under the static-retention bias. Figures 4.5(a) and 4.5(b) show the resulting V_{SN} evolution for the ‘1’ and ‘0’ states, and Figure 4.6 illustrates the corresponding energy-band diagrams under this WL-off, $BL = V_{DD}/2$ condition.

In the ‘1’ state, the BL side serves as the effective drain, making the BL-adjacent junction the dominant leakage path. Figure 4.5(a) shows that the configuration with a high-WF metal on the BL side and a low-WF metal on the SN side ($M_1/M_2 = 4.9/4.5$ eV) yields the smallest V_{SN} drop. The band diagram in Figure 4.6(a), extracted under the same static-retention bias, confirms that the high-WF BL-side gate raises the local channel barrier and weakens the BL-side electric field, thereby suppressing leakage from the channel toward the BL. When the polarity is reversed, and the BL-side gate uses a low work function ($M_1 = 4.5$ eV), the junction barrier becomes shallower and the leakage increases, resulting in the largest V_{SN} degradation in the ‘1’ state.

In the ‘0’ state, the source and drain roles swap under WL-off and $BL = V_{DD}/2$, so the SN side becomes the primary leakage path. Figure 4.5(b) shows that the opposite WF polarity—low-WF M_1 and high-WF M_2 (4.5/4.9 eV)—provides the most stable retention, as the high-WF SN-side metal strengthens the local barrier and suppresses leakage toward the SN, consistent with the band profile in Figure 4.6(b). Conversely, assigning a low work function to the SN-side gate reduces this barrier and leads to a larger V_{SN} rise; however, the high-WF BL-side metal raises the local barrier and suppresses leakage toward the BL junction.

These tendencies are summarized in Figure 4.5(c): both asymmetric combinations outperform the symmetric 4.7/4.7 eV gate, but the optimal WF polarity differs for the ‘1’ and ‘0’ states because each state is governed by a different dominant leakage path. Overall, the dual-WF gate improves static retention not by reducing FBE itself, but by reshaping junction barriers under the static-retention bias so that the relevant leakage path is minimized in each state.



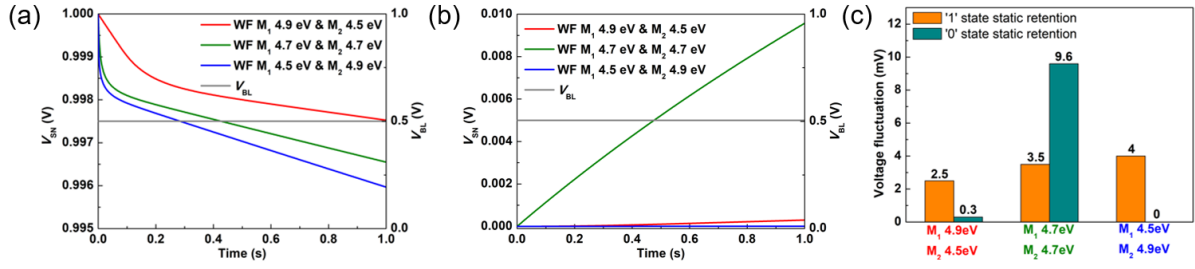


Figure 4.5. Static-retention characteristics of the dual work-function (dual-WF) vertical-channel transistor for various work-function pairs ($M_1/M_2 = 4.9/4.5, 4.7/4.7, 4.5/4.9$ eV). (a) Temporal evolution of the storage-node (SN) voltage for the '1' state under WL-off and $BL = V_{DD}/2$ bias. (b) Corresponding SN-voltage variation for the '0' state. (c) Total SN-voltage fluctuation after 1 s of retention.

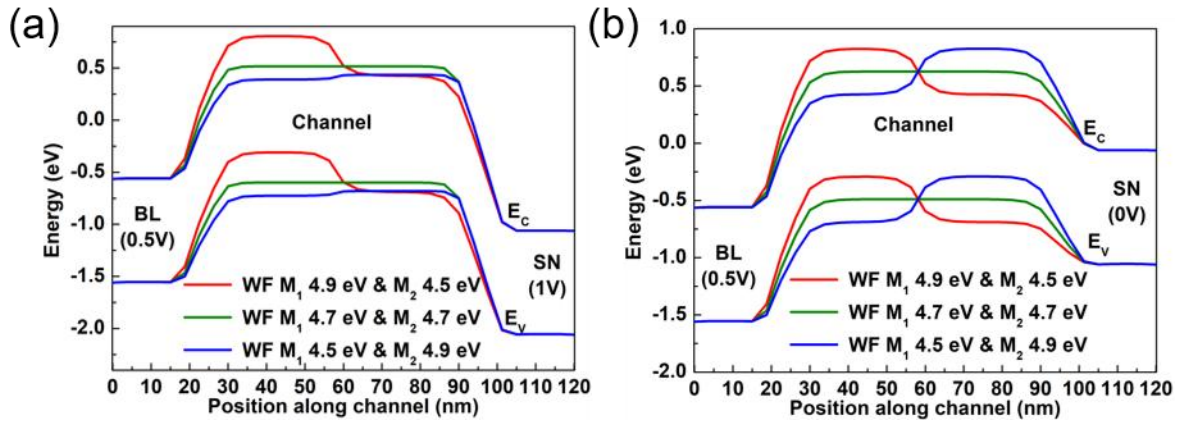


Figure 4.6. Energy-band diagrams of the dual-WF vertical transistor under static-retention bias (WL off, $BL = V_{DD}/2$). (a) Band profile for the '1' state. (b) Band profile for the '0' state.

4.2.4 Evaluation of Pure Floating-Body Effect

Figure 4.7 quantifies the instantaneous floating body charging during a single BL-high interval by examining the valence-band energy profile at two time points (t_1, t_2) and the resulting BPV. Unlike retention, which depends on cumulative charging and discharging over multiple BL toggles, BPV captures the instantaneous hole generation determined by local junction fields.

In the '1' state, Figure 4.7(a) shows how the valence-band edge evolves during the BL-high interval. The BPV trend in Figure 4.7(c) follows the BL-side electric-field strength determined by the BL-side work function (M_1). The high-WF M_1 / low-WF M_2 (4.9/4.5 eV) device produces the largest BPV because the elevated BL-side work function strengthens the BL-adjacent electric field and maximizes GIDL-driven hole generation. The

single-WF (4.7/4.7 eV) configuration exhibits an intermediate BL-side field and a moderate BPV. In contrast, the low-WF M_1 / high-WF M_2 (4.5/4.9 eV) device weakens the BL-side field and yields the smallest BPV despite its poorer retention performance, reflecting minimal instantaneous hole generation during BL-high.

In the '0' state, the source and drain roles reverse, and the BL side becomes the drain under the BL-high bias condition. Figure 4.7(b) shows the corresponding band profiles in this state. The BPV trend in Figure 4.7(c) also reverses: the high-WF M_1 / low-WF M_2 (4.9/4.5 eV) device lowers the BL-adjacent barrier and enhances hole generation at the BL junction, resulting in the largest BPV. Meanwhile, the low-WF M_1 / high-WF M_2 (4.5/4.9 eV) configuration raises the BL-adjacent barrier, suppresses GIDL, and yields the smallest BPV. The symmetric 4.7/4.7 eV device maintains relatively strong fields at both junctions and therefore exhibits comparatively large BPV.

These observations confirm that Figure 4.7(a) captures BL-side hole generation in the '1' state, Figure 4.7(b) captures SN-side hole generation in the '0' state, and Figure 4.7(c) summarizes the resulting BPV differences across work-function combinations. Importantly, since BPV reflects only instantaneous body charging during a single BL-high event, it does not directly predict dynamic-retention performance. Retention degradation instead depends on the cumulative balance between charging and discharging over repeated toggling, explaining why the configuration with the lowest BPV does not necessarily show the best retention.

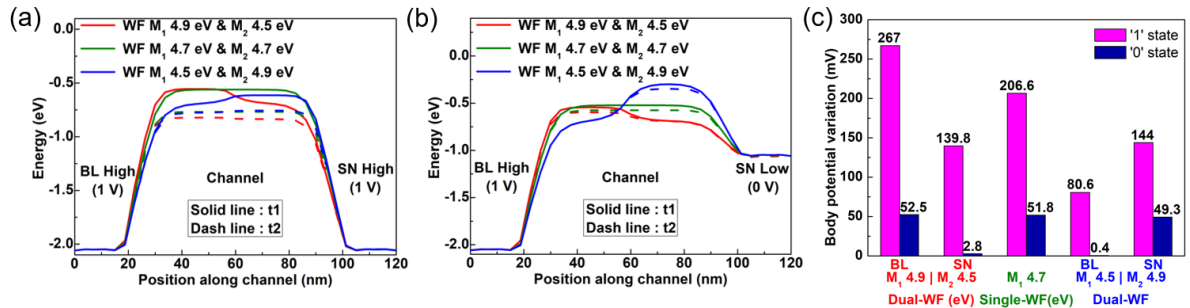
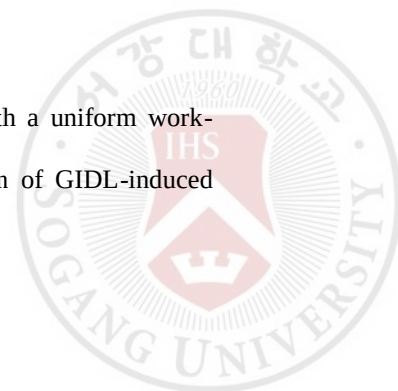


Figure 4.7. BPV characteristics of dual-WF devices. (a) Valence-band profiles during BL-high for the '1' state. (b) Valence-band profiles during BL-high for the '0' state. (c) Extracted BPV for both storage states across different WF combinations.

4.2.5 Discussion and Summary

The dual-WF gate provides localized electrostatic control that cannot be achieved with a uniform work-function gate, enabling selective suppression of junction fields and measurable reduction of GIDL-induced



floatingbody charging. The analysis across dynamic retention, static retention, and BPV extraction consistently shows that adjusting the work-function polarity reshapes the BL- and SN-side barriers in opposite ways, leading to improved stability in one storage state while inevitably degrading the other. This state-dependent behavior originates from the fact that the BL and SN junctions alternately act as the dominant conduction or leakage paths depending on the logic state and bias condition.

The BPV results further highlight that instantaneous body charging during BL-high does not directly predict long-term retention stability: the polarity that minimizes BPV may still suffer from degraded retention due to unfavorable barrier conditions during BL-low. Conversely, the polarity that enhances retention can exhibit larger BPV because field relaxation and barrier modulation influence the two phases differently. These observations confirm that a two-segment dual-WF structure, despite offering meaningful field control, is fundamentally constrained by the opposing electrostatic requirements of the ‘1’ and ‘0’ states.

Thus, while dual-WF engineering marks a significant improvement over a single-WF gate, its inherent two-segment configuration cannot simultaneously meet the stability criteria for both storage states. This limitation motivates the introduction of a three-segment work-function structure, which provides an additional degree of freedom to independently tune the source- and drain-adjacent junction fields, enabling balanced suppression of FBE across both logic states.

4.3 Three-Layer Work-Function Gate Engineering for Balanced FBE Mitigation

4.3.1 Introduction

To overcome the dual-WF trade-off, we propose a three-layer work-function (3-layer WF) gate, as illustrated in Figure 4.8. The gate electrode is vertically segmented into Metal 1 (low WF, BL side), Metal 2 (high WF, center), and Metal 3 (low WF, SN side). The low-WF end segments relax the local gate–drain fields at both channel ends, suppressing GIDL-driven body charging, while the high-WF middle segment enhances channel depletion and limits leakage. The result is a balanced dynamic and static retention under a single WL.



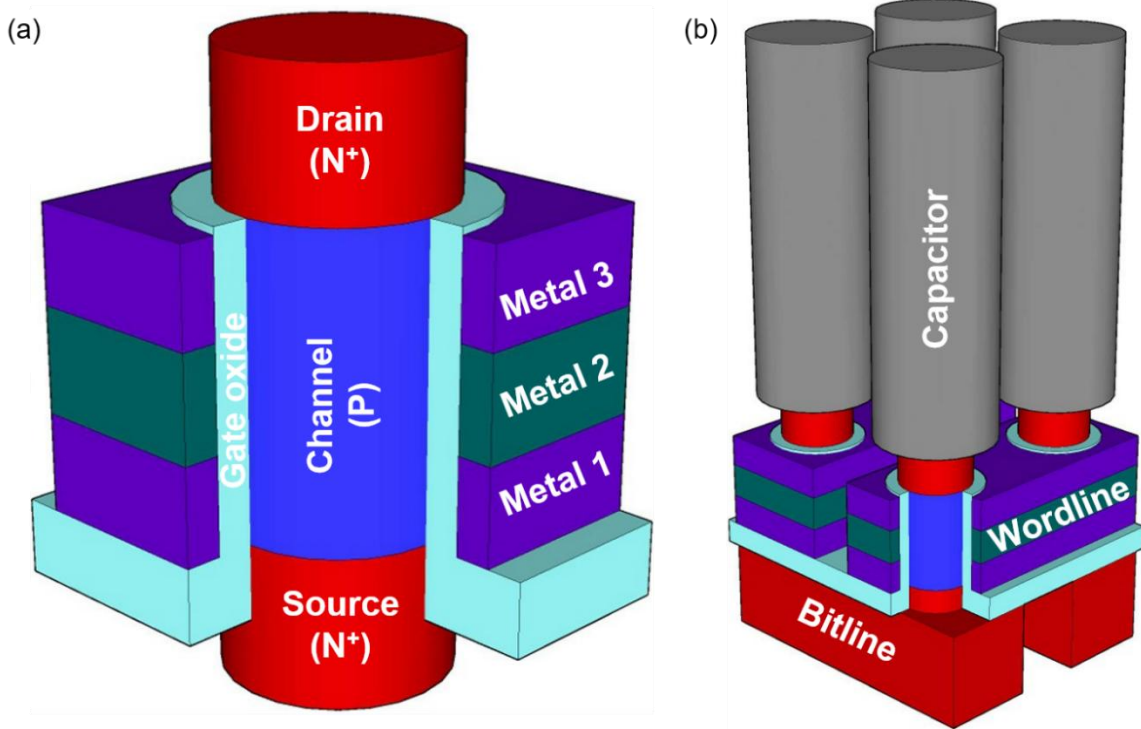
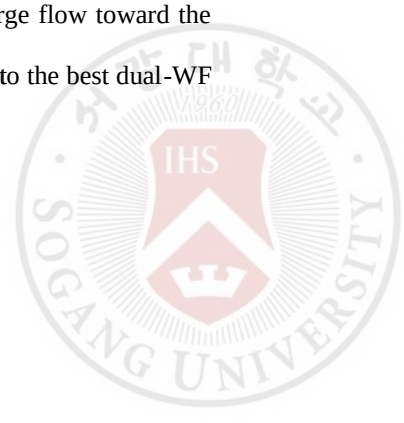


Figure 4.8. (a) 3D schematic of the proposed vertical-channel transistor employing a three-layer work-function (3-layer WF) gate. (b) Conceptual array-level configuration of the 3-layer WF vertical DRAM cell.

4.3.2 Dynamic Retention Characteristics

The dynamic-retention characteristics of the 3-layer WF device were examined by comparing its V_{SN} evolution with dual-WF references. In the '1' state, the waveform in Figure 4.9(a) shows that the 3-layer configuration exhibits the smallest V_{SN} drop during BL toggling. This improvement originates from the combined effect of the low-WF outer segments (M_1 and M_3), which weaken the BL- and SN-adjacent junction fields, and the high-WF center segment (M_2), which maintains strong channel depletion. As a result, GIDL is simultaneously suppressed at both junctions, yielding the lowest voltage fluctuation after ten cycles, as summarized in Figure 4.9(c).

When the bias polarity is reversed in the '0' state, the BL terminal becomes the active drain. The V_{SN} trend in Figure 4.9(b) indicates that the 3-layer WF device again limits BL-side GIDL through its low-WF BL-adjacent segment (M_1), while the high-WF center segment preserves a robust barrier that slows charge flow toward the SN. Although the resulting V_{SN} rise is not the absolute minimum, it remains nearly identical to the best dual-WF configuration, demonstrating resilient retention even under the reversed bias condition.



Further insight into these trends is provided by the band-edge profiles in Figure 4.10. Figure 4.10(a) illustrates the valence-band evolution during the ‘1’ state, where both the BL and SN junction fields are relaxed by the low-WF outer metals, and the central high-WF segment maintains channel depletion—consistent with the reduced body charging observed in Figure 4.9(a). Figure 4.10(b) shows the corresponding profile for the ‘0’ state. Under this condition, the BL-side barrier becomes the dominant factor, and the low-WF BL-adjacent segment reduces BL-side GIDL while the high-WF central segment prevents excessive discharge toward the SN. This band behavior aligns with the moderated V_{SN} rise observed in Figure 4.9(b).

Taken together, the voltage and band-profile results indicate that the 3-layer WF structure mitigates junction-specific GIDL for both logic states and alleviates the state-dependent trade-off inherent to dual-WF designs.

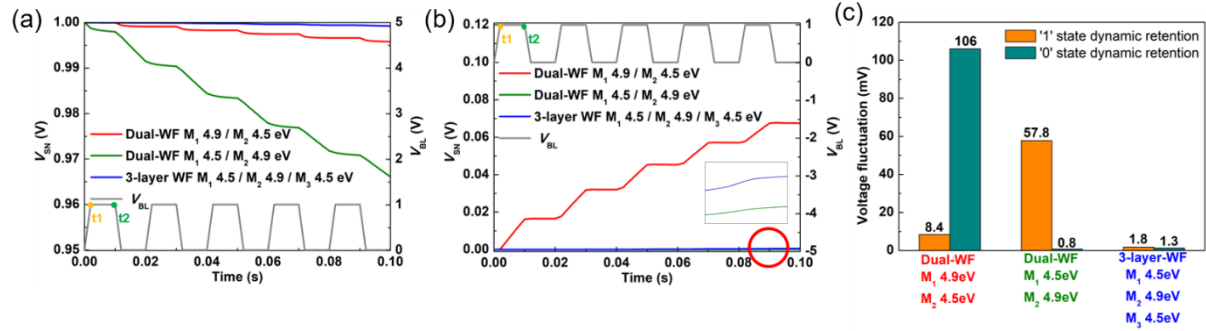


Figure 4.9. Dynamic-retention characteristics of dual-WF and 3-layer WF vertical transistors. (a) Transient SN-voltage for the ‘1’ state. (b) Transient SN-voltage for the ‘0’ state. (c) SN-voltage fluctuation after ten BL-toggling cycles.

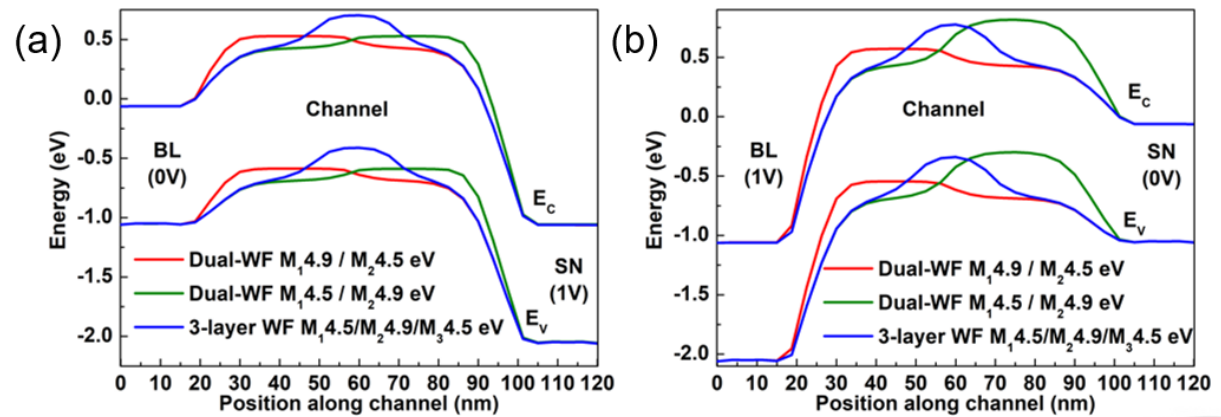
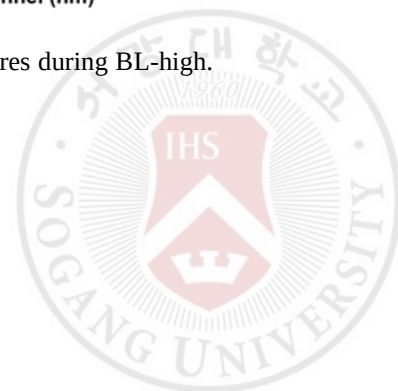


Figure 4.10. Energy-band diagrams comparing the 3-layer WF device with dual-WF structures during BL-high. (a) Band profile for the ‘1’ state. (b) Band profile for the ‘0’ state.



4.3.3 Static Retention Characteristics

The static-retention characteristics under WL-off and $BL = V_{DD}/2$ reveal that the 3-layer WF architecture provides the most balanced suppression of steady-state degradation across both storage states.

For the '1' state, the SN-voltage decay in Figure 4.11(a) shows that the 3-layer WF device exhibits the smallest V_{SN} drop. This improvement stems from the combined effect of the low-WF outer metals (M_1/M_3), which relax junction fields and reduce GIDL-driven charging at both terminals, and the high-WF center metal (M_2), which maintains strong channel depletion and suppresses steady-state leakage. The energy-band profile in Figure 4.12(a) confirms that the 3-layer WF produces both lower junction fields and a reinforced channel barrier, leading to improved retention.

For the '0' state, Figure 4.11(b) indicates that the 3-layer WF yields a V_{SN} rise comparable to the best-performing dual-WF configuration (4.5/4.9 eV). The band profile in Figure 4.12(b) explains this behavior: the low-WF BL segment (M_1) weakens the BL-adjacent field and suppresses GIDL, while the high-WF center segment (M_2) prevents excessive lowering of the SN-side barrier, thereby limiting discharge imbalance.

An additional trend emerges when comparing the 3-layer WF to the dual-WF 4.9/4.5 eV configuration. Although the dual-WF 4.9/4.5 device initially shows a smaller V_{SN} increase, Figure 4.11(b) reveals that its degradation accelerates over time. This is consistent with the higher local barrier observed in Figure 4.12(b): the elevated BL-side work function raises the valence-band barrier, but because GIDL occurs at this same BL junction, generated holes gradually lower the barrier during stress. As the barrier collapses, leakage increases progressively, resulting in worse long-term retention than the 3-layer WF case.

The overall static-retention comparison in Figure 4.11(c) highlights this effect: the 3-layer WF delivers the best V_{SN} stability in the '1' state and maintains competitive performance in the '0' state, avoiding the polarity-dependent limitations of dual-WF designs. By providing simultaneous BL/SN-side GIDL suppression together with strong channel depletion, the 3-layer WF achieves the most robust and balanced static-retention behavior among the evaluated configurations.



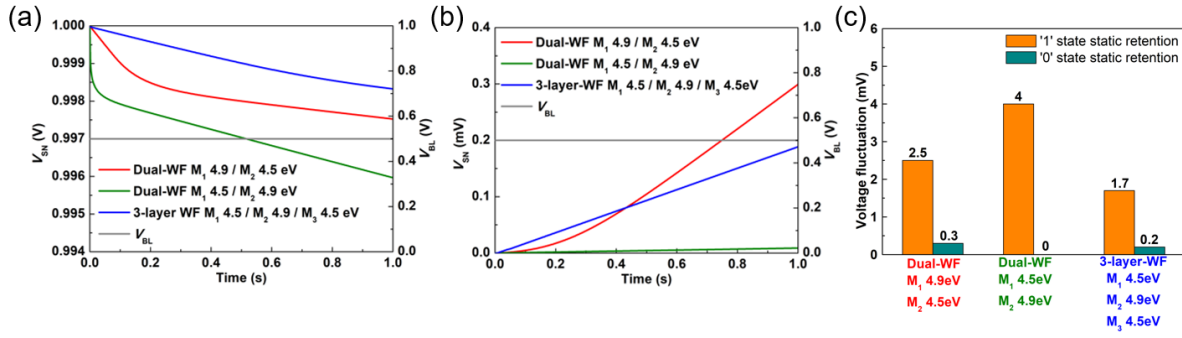


Figure 4.11. Static-retention characteristics of dual-WF and 3-layer WF vertical transistors. (a) SN-voltage variation for the '1' state. (b) SN-voltage variation for the '0' state. (c) SN-voltage fluctuation after 1 s retention.

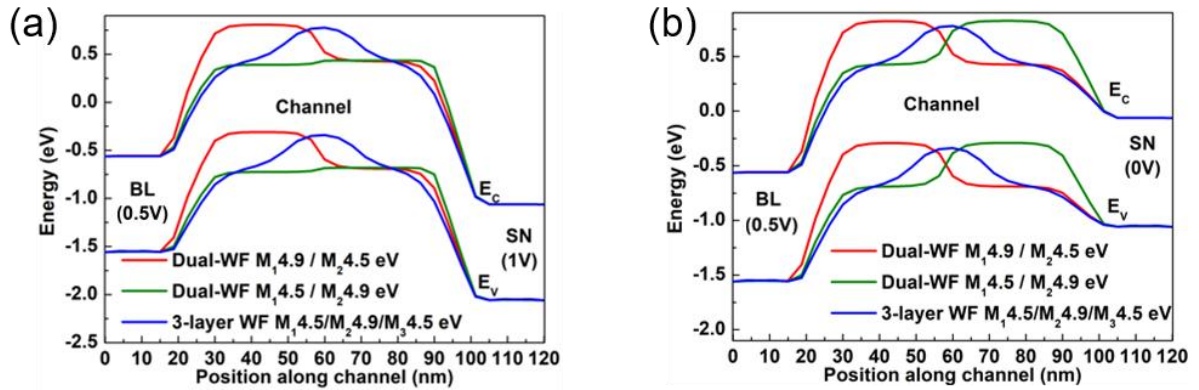


Figure 4.12. Static-retention characteristics of dual-WF and 3-layer WF vertical transistors. (a) SN-voltage variation for the '1' state. (b) SN-voltage variation for the '0' state. (c) SN-voltage fluctuation after 1 s retention.

4.3.4 Evaluation of Pure Floating-Body Effect

The instantaneous floating body behavior was quantified by extracting the BPV during the BL-high phase, as shown in Fig. 4.13. Figures 4.13(a) and 4.13(b) present the valence-band evolution at two sampling instants (t_1 and t_2) for the '1' and '0' states, respectively, and Fig. 4.13(c) summarizes the resulting BPV magnitudes.

In the '1' state, the dual-WF devices exhibit large BPV because only one junction field is relaxed: the 4.9/4.5 eV case reduces the SN-side field but leaves the BL-side field strong, whereas the 4.5/4.9 eV case relaxes the BL side but enhances the SN-side field. As reflected in Fig. 4.13(a), both configurations generate substantial hole injection at one side of the body, producing large BPV. In contrast, the 3-layer WF structure (low-WF M_1 / high-WF M_2 / low-WF M_3) simultaneously moderates the BL and SN junction fields while maintaining strong

channel depletion through the center high-WF segment. This balanced field distribution suppresses GIDL at both ends and limits the amount of stored charge, resulting in the smallest BPV among all cases.

A similar trend appears in the '0' state, where the junction roles are reversed. As shown in Fig. 4.13(b), each dual-WF device again relaxes only one junction field and thus permits sizable hole generation at the opposite junction. The 3-layer WF device continues to suppress GIDL at both ends, lowering the overall BPV even when the bias polarity is reversed.

Because BPV reflects the instantaneous charging capability of each WF configuration, the results in Fig. 4.13(c) clearly demonstrate that the 3-layer WF architecture provides superior suppression of the pure FBE in both logic states. By reshaping the longitudinal electric field with two low-WF ends and a high-WF center, the 3-layer WF gate minimizes hole generation and prevents deep body charging, outperforming all dual-WF references.

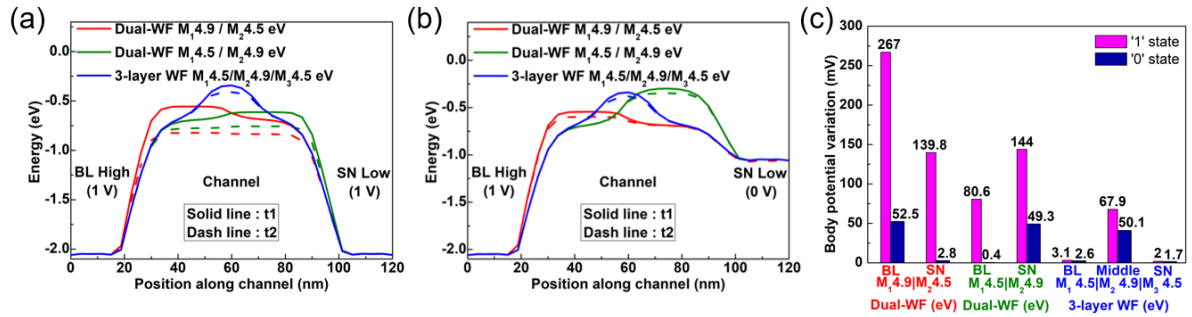


Figure 4.13. BPV characteristics of dual-WF and 3-layer WF devices. (a) Valence-band profiles during BL-high for the '1' state. (b) Valence-band profiles during BL-high for the '0' state. (c) Extracted BPV for both storage states, comparing dual-WF and 3-layer WF configurations.

4.3.5 Fabrication Feasibility

In addition to its electrical effectiveness, the proposed three-layer work-function (3-layer WF) gate structure can be implemented in a pillar-type vertical-channel transistor (VCT) using a fabrication flow that is fully compatible with conventional DRAM processes. Because the gate segmentation is defined along the vertical axis of the pillar, the 3-layer WF gate can be realized through sequential deposition and etch-back steps without any additional photolithography or lateral patterning.



The key modules required to realize a $4F^2$ vertical-pillar access transistor (VPT) employing a three-layer work-function gate segmentation—designed to provide fine electrostatic control along the vertical channel and to suppress FBE—are summarized in Figure 4.14.

The process begins with the formation of a vertically stratified $N^+/P/N^+$ silicon stack that predefines the bottom N^+ source, the p-type channel body, and the top N^+ drain, as illustrated in Figure 4.14(a). The stack can be fabricated by a three-step chain implantation or a selective epitaxial growth sequence followed by dopant activation.

A silicon pillar is then patterned by anisotropic etching through the multilayer stack and stopped within the source region, as shown in Figure 4.14(b). A conformal inter-layer dielectric (ILD) is deposited and planarized to provide lateral isolation while maintaining contact access to the bottom source pad, as shown in Figure 4.14(c).

Subsequently, an all-around gate dielectric is deposited by ALD to guarantee uniform coverage of the pillar surface, as shown in Figure 4.14(d). The 3-layer metal gate stack is then formed by sequential metal deposition of three work-function-tuned metals (M_1 , M_2 , M_3), each corresponding to the drain-proximal, central-body, and source-proximal regions of the vertical channel, as shown in Figure 4.14(e). This stacked-metal segmentation provides localized control of the channel potential without requiring additional lithography steps.

Finally, the drain pad is opened through a selective etch and connected to the storage-node (SN) capacitor formed above the pillar, while the bitline (BL) is routed beneath and contacted to the source, as depicted in Figure 4.14(f). The resulting structure completes the vertical separation of BL and SN and achieves $4F^2$ array tiling with no lateral-area penalty, enabling robust channel-potential modulation and improved stability against floating body-induced variability.

In summary, the 3-layer WF structure offers not only balanced suppression of the FBE but also superior manufacturability. Its vertically stacked gate metals can be deposited and patterned through standard thin-film modules, eliminating the need for extra masks or complex multi-WL routing, and thus providing a highly practical path toward next-generation vertical DRAM integration.



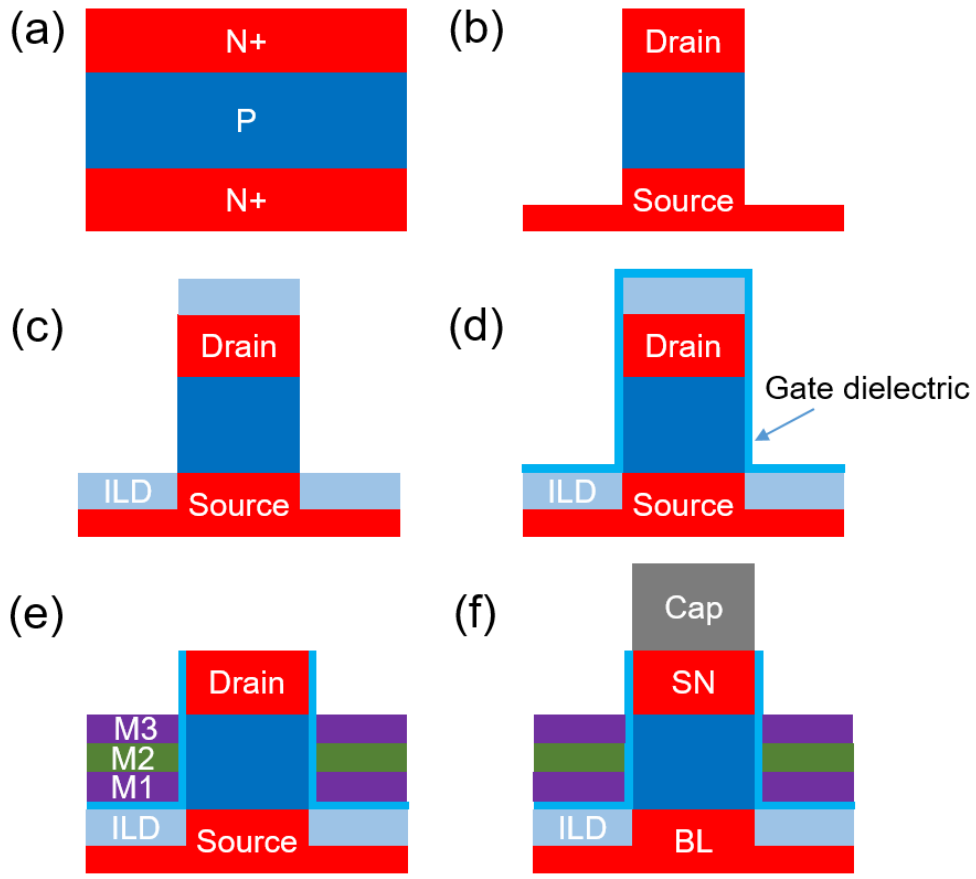
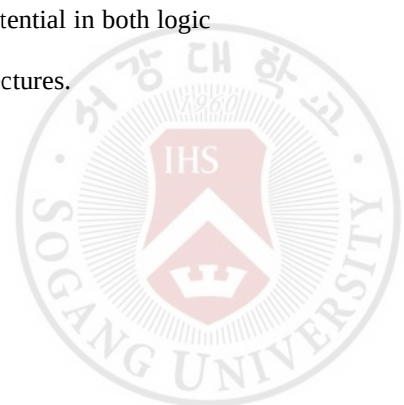


Figure 4.14. Fabrication sequence of the proposed three-layer work-function (3-layer WF) vertical-channel transistor (VCT) compatible with DRAM processes. (a) Formation of a vertically stratified N⁺/P/N⁺ silicon stack defining the source, channel, and drain regions. (b) Pillar patterning by anisotropic etching through the multilayer stack. (c) Deposition and planarization of inter-layer dielectric (ILD) for lateral isolation and source-pad access. (d) Conformal deposition of all-around gate dielectric by ALD. (e) Sequential deposition of three work-function-tuned gate metals (M₁, M₂, M₃) along the vertical pillar to form the 3-layer WF gate stack. (f) Final integration of the storage-node (SN) capacitor on the drain side and bit-line (BL) contact to the source, completing the 4F² vertical DRAM cell structure.

4.3.6 Discussion and Summary

The proposed 3-layer WF gate effectively resolves the intrinsic trade-off encountered in dual-WF designs by introducing a third degree of electrostatic control along the vertical channel. Through spatial segmentation of the gate work function—implementing low-WF metals at the channel ends and a high-WF metal at the center—the device simultaneously weakens the gate-induced drain leakage (GIDL) field while enhancing depletion in the central region. This engineered longitudinal field distribution stabilizes the floating body potential in both logic states and mitigates asymmetric hole generation, a limitation inherent to two-segment architectures.



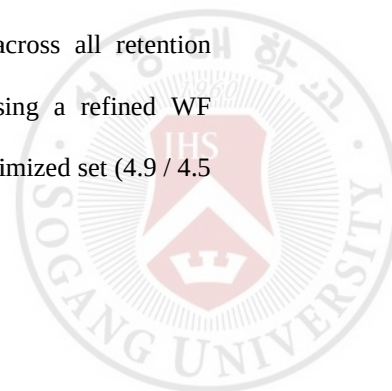
Dynamic retention analysis confirms that the 3-layer WF structure yields the smallest V_{SN} degradation in the ‘1’ state among all dual-WF references. This improvement results from the cooperative field modulation provided by the SN-side low WF, which relaxes the drain-side electric field, and the central high WF, which sustains body depletion. In the ‘0’ state, the 3-layer configuration maintains retention performance comparable to the best dual-WF design, demonstrating that the additional WF segment effectively suppresses the polarity dependence observed in dual-WF devices. The extracted body-potential variation (BPV) further validates that the proposed structure limits instantaneous hole generation and significantly reduces body-potential excursions over multiple BL toggling cycles.

The benchmark comparison in Table 4.1 highlights the broader significance of these improvements relative to prior FBE-mitigation approaches [4.4], [4.6]-[4.8]. Earlier studies have reported enhancements primarily for the ‘1’-state dynamic retention, while data for the ‘0’ state and static retention were generally absent or degraded. As such, these designs did not fully address the long-term floating-body instability spanning both logic polarities.

Reference	[4.4]	[4.6]	[4.7]	[4.8]	This Work
Engineering Method	Multi-WL	LDD	Inner Gate	Dual-WF	Three-layer WF
‘1’ State Dynamic Retention	97.5 %	78 %	~ 25 %	85 %	99.3 %
‘0’ State Dynamic Retention	N/R	N/R	N/R	N/R	99.5 %
‘1’ State Static Retention	N/R	N/R	Worse	80.8 %	85 %
‘0’ State Static Retention	N/R	N/R	N/R	N/R	99.6 %

Table 4.1. Benchmark comparison of prior floating-body-effect (FBE) mitigation techniques and the proposed three-layer work-function (WF) gate.

In contrast, the proposed 3-layer WF architecture was evaluated comprehensively across all retention conditions: both ‘1’ and ‘0’ states, and both dynamic and static retention modes. Using a refined WF combination (4.95 / 4.45 / 4.95 eV), which provides results consistent with the originally optimized set (4.9 / 4.5



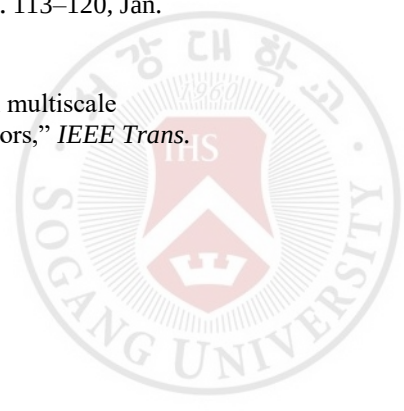
/ 4.9 eV), the 3-layer WF device achieved the highest performance in every category. These results confirm that the proposed field-profile engineering strategy not only suppresses short-term hole bursts but also minimizes long-term body drift, yielding balanced and robust retention behavior beyond what has been demonstrated in prior work.

Beyond electrical performance, the manufacturability of the 3-layer WF gate represents a key advantage. Because all WF segments are introduced through vertical stacking using standard deposition and etch-back steps, the configuration avoids the routing, overlay, and lithography penalties associated with multi-WL schemes. This process flow is fully compatible with existing high-k/metal-gate (HKMG) modules, facilitating seamless integration into contemporary DRAM fabrication.

Overall, the 3-layer WF gate provides a practical and scalable solution for mitigating the floating-body effect (FBE) in vertical DRAM access transistors. By reshaping the longitudinal electric field rather than relying solely on geometric or doping-based scaling, the design achieves strong and polarity-independent retention characteristics without compromising drive current or process simplicity. These findings underscore field-profile engineering as a critical pathway toward reliable, high-density next-generation DRAM technologies.

4.4 Reference

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Chapter 5. Conclusion

This work presented a unified framework for analyzing and mitigating the FBE in vertical-channel DRAM access transistors. Using a rigorously calibrated 3D mixed-mode TCAD environment, we linked FBE-driven body-potential excursions to dynamic and static retention loss, and introduced BPV and duty-controlled dynamic-retention metrics to distinguish instantaneous charging from cycle-averaged degradation. Scaling studies revealed that shorter gate length strengthens drain-field coupling and increases GIDL-driven hole generation, while thinner body improves electrostatic control until quantum-confinement effects (QCE) raise the conduction-band edge and reduce inversion charge, defining a practical T_{Body} design window. Drain doping offers limited leverage—suppressing GIDL but penalizing On-current and access speed.

Building on these insights, we evaluated field-engineering approaches that retain a single WL while relaxing the drain-adjacent field. Prior split-gate work confirms that auxiliary-gate bias can suppress GIDL but introduces routing and process overhead. Our dual work-function (dual-WF) gate reproduces the field-relaxation benefit using two stacked metals, improving ‘1’-state retention and reducing BPV, yet exhibits a polarity-dependent trade-off because the ‘1’ and ‘0’ states require opposite barrier configurations. To overcome this asymmetry, we proposed a three-layer work-function (3-layer WF) gate that places low-WF metals at both channel ends and a high-WF segment in the center. This reshapes the longitudinal field profile so that junction fields are simultaneously softened while channel depletion is preserved, minimizing body-potential excursions and providing balanced retention across states. The structure can be fabricated along the pillar axis by sequential deposition and etch-back without additional lithography, remaining compatible with DRAM HKMG processing.

Overall, this dissertation reframes FBE as a controllable design parameter. Through calibrated device/circuit co-simulation, physically grounded FBE metrics, and longitudinal WF segmentation, we provide actionable guidelines: select gate length and body thickness within the QCE-aware window; avoid excessive drain doping; and employ work-function segmentation—preferably the 3-layer WF—to decouple state-dependent constraints. These results outline a practical path toward robust, high-density vertical DRAM and extend naturally to memory and compute-in-memory platforms influenced by floating body behavior.



Summary

Floating-Body Effect Analysis and Multi-Layer Gate Engineering for Vertical-Channel DRAM Access Transistors

본 연구는 차세대 고집적 DRAM 에서 핵심 소자인 수직 채널 접근 트랜지스터(vertical-channel transistor, VCT)에서 발생하는 부유체 효과(Floating-Body Effect, FBE)의 기원과 스케일링 의존성을 규명하고, 이를 억제하기 위한 전계 기반(field-profile engineering) 게이트 구조 설계를 제안한다. 수직형 DRAM 구조에서는 실리콘 채널이 주변 절연막으로 부분적으로 고립되어 있어, GIDL·BTBT 로 생성된 홀이 채널 내부에 축적되고 이는 체적 전위(body potential)를 변동시켜 데이터 유지(retention)를 제한하는 근본적인 신뢰성 문제로 이어진다.

본 논문에서는 실측 기반 3D 혼합모드 TCAD 를 활용하여 FBE 의 물리 메커니즘을 정량적으로 분석하였다. 게이트 길이(L_G)가 짧아질수록 드레인 전계 결합(drain-field coupling)이 강화되어 GIDL 기반 홀 생성이 증가하고, 바디 두께(T_{Body})가 얇아질수록 게이트 제어력이 향상되어 FBE 가 억제되는 반면, 수 nm 이하에서는 양자 구속 효과(QCE)가 밴드 에지 상승 및 구동 전류 감소를 유발하여 전류 성능과의 trade-off 가 발생함을 보였다. 분석 결과, FBE 억제와 구동 전류 확보 간의 균형점은 $T_{Body} \approx 5$ nm 구간에서 형성됨을 확인하였다. 또한 드레인(SN) 도핑 농도는 GIDL 억제에는 효과적이지만 과도한 저감 시 소스 저항 증가로 인해 접근 속도를 저하시켜, 단순 도핑 조정만으로는 FBE 문제를 해결할 수 없음을 밝혔다.

이를 기반으로 본 연구는 단일 워드라인 구조를 유지하면서 전계 프로파일을 공학적으로 조절할 수 있는 다층 일함수 게이트(multi-layer work-function gate) 구조를 제시하였다. 우선, 듀얼 일함수(dual-WF) 구조는 단일 게이트 구조 대비 전계 조절이 가능하나 '1' / '0' 상태에서 요구되는 장벽 조건이 서로 반대여서 구조적으로 필연적인 polarity trade-off 가 발생함을 명확히 규명하였다. 특히, 본 논문에서는 Body-Potential Variation (BPV)와 dynamic retention 의 관계를 구분하여, BPV 최소 조건이 retention 최적 조건과 반드시 일치하지 않는다는 점—즉 BL-side GIDL, discharge 통로, junction polarity 에 따라 순간(body charging)과 누적(retention) 메커니즘이 상이함—을 정량적으로 규명하였다.

이러한 dual-WF 한계를 극복하기 위해 3-층 일함수(3-layer WF) 게이트 구조를 제안하였다. 채널의 양단은 low-WF 를 배치하여 BL·SN 양측 junction field 를 동시에 완화하고, 중심부는 high-WF 로 유지하여 채널 전하를 안정화함으로써, 두 논리 상태('1' · '0') 모두에서 균형 잡힌 FBE 억제를 실현한다. 그 결과, 3-layer WF 구조는 dual-WF 에서 나타나는 편극 의존성을 제거하고, 두 논리 상태 모두에서 가장 작은 BPV 와 안정적인 retention 특성을 달성하였다. 또한 세 금속층은 수직 적층과 단계적 etch-back 만으로 구현할 수 있어, 추가 마스크나 복잡한 패터닝 없이 기존 DRAM HKMG 공정과 자연스럽게 호환되는 제조 용이성을 갖는다.

종합하면, 본 연구는 FBE 를 단순한 scaling 한계가 아니라 전계 기반 설계 변수로 재정의하였으며, 게이트 길이·바디 두께·도핑·일함수 분할을 아우르는 체계적 분석과 구조적 해결책을 제시하였다. 이는 차세대 수직 DRAM 메모리 구조의 신뢰성·성능·제조 가능성을 동시에 향상시킬 수 있는 현실적이고 확장 가능한 설계 지침으로 활용될 수 있다.



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무엇보다도, 가족들의 믿음과 존중이 없었다면 대학원 진학과 석사 과정은 시작조차 할 수 없었을 것입니다. 결혼 직후 회사를 그만두고 대학원에 진학하겠다는 제 선택을 이해해주고 묵묵히 함께해준 사랑하시는 아내 강민지에게 깊이 감사하며, 진심으로 사랑합니다. 당신의 희생이 헛되지 않도록, 대학원 과정에서 배운 지식과 경험을 바탕으로 더욱 성장해 나가는 남편이 되겠습니다. 항상 제 대학원 생활에 관심을 가져주시고, 당신의 경험을 바탕으로 아낌없는 조언을 해주신 아버지께도 감사드립니다. 과거에 직장 생활과 가족을 책임지시며 박사 학위를 받으셨던 아버지의 노력이 얼마나 큰 것이었는지를 이제서야 조금이나마 이해하게 되었고, 그 점에서 진심으로 존경의 마음을 전합니다. 언제나 제 편이 되어주시고 따뜻하게 응원해주신 어머니께도 감사드리며, 앞으로는 더욱 살가운 아들이 되도록 노력하겠습니다. 또한 저처럼 방황의 시기를 보내고 있는 동생 의준에게, 자신의 방향을 찾아 더 행복한 앞날을 맞이하길 진심으로 응원합니다.

서강대학교 전자공학과 대학원에 입학한 것은 제 인생에서 큰 행운이었습니다. 훌륭한 교수님들로부터 가르침과 지도를 받을 수 있었음에 깊이 감사드립니다. 먼저, 책임감을 가지고 끝까지 논문 지도를 맡아주신 한준규 교수님께 진심으로 감사드립니다. 학문적 지식뿐만 아니라 연구에 임하는 자세와 끊임없이 노력하는 태도까지 많은 것을 배울 수 있었습니다. 또한 논문 전반에 걸쳐 관심을 가지고 지도해주시고, 여러 행정적인 부분에서도 도움을 주신 김시현 교수님께도 깊이 감사드립니다. 아울러 논문 심사를 맡아주신 김상완 교수님과 김장생 교수님께도 감사의 말씀을 드립니다. 교수님들의 소중한 조언과 관심 덕분에 연구를 보완하고 학위를 마무리할 수 있었습니다.

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대학원 과정은 저에게 학문적 지식을 습득하는 시간에 그치지 않고, 문제에 직면했을 때 이를 해결해 나가는 방법과 원하는 결과를 얻기까지 쉽게 포기하지 않고 노력하는 태도를 배우는 과정이었습니다. 또한 연구뿐만 아니라 삶의 여러 영역에서 모든 일이 항상 제 뜻대로 흘러가지는 않는다는 사실을 받아들이며, 스스로를 돌아보는 겸손함의 중요성도 깨닫게 되었습니다. 이 과정에서 배운 학문적 지식과 삶의 태도를 소중히 간직하여, 앞으로의 연구와 삶에서도 끊임없이 성장해 나가는 사람이 되도록 노력하겠습니다. 이러한 값진 경험을 할 수 있었음에 깊이 감사드립니다.

